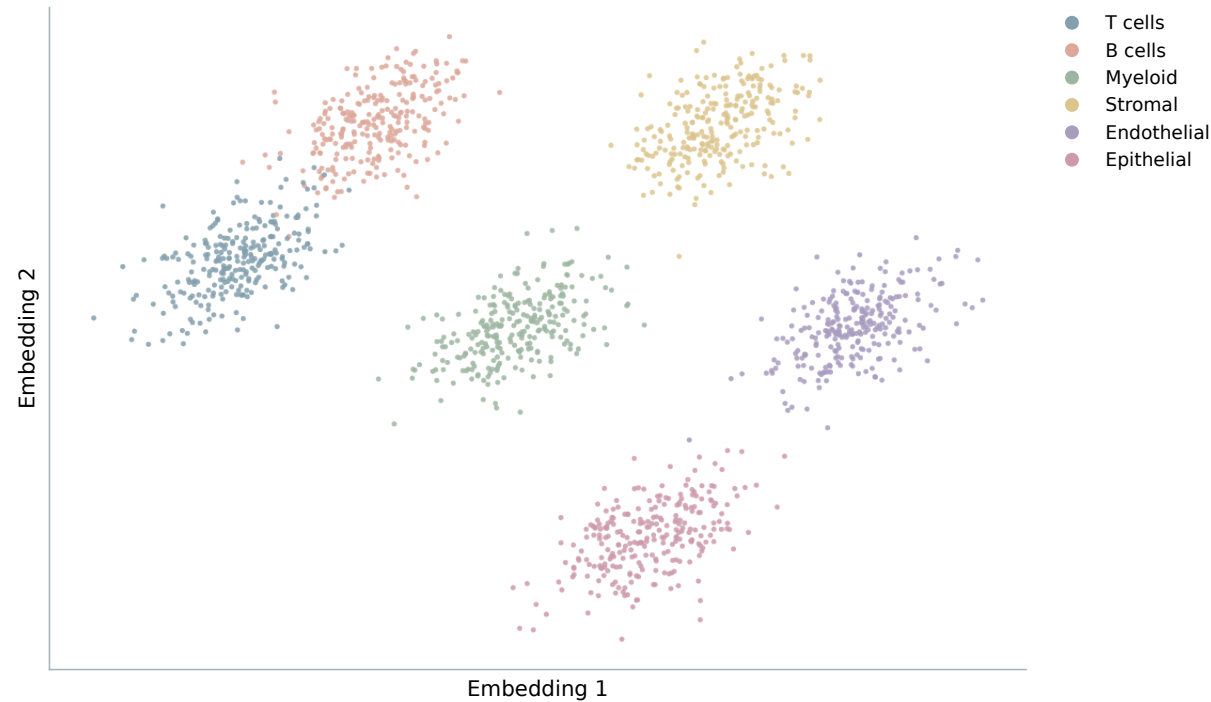
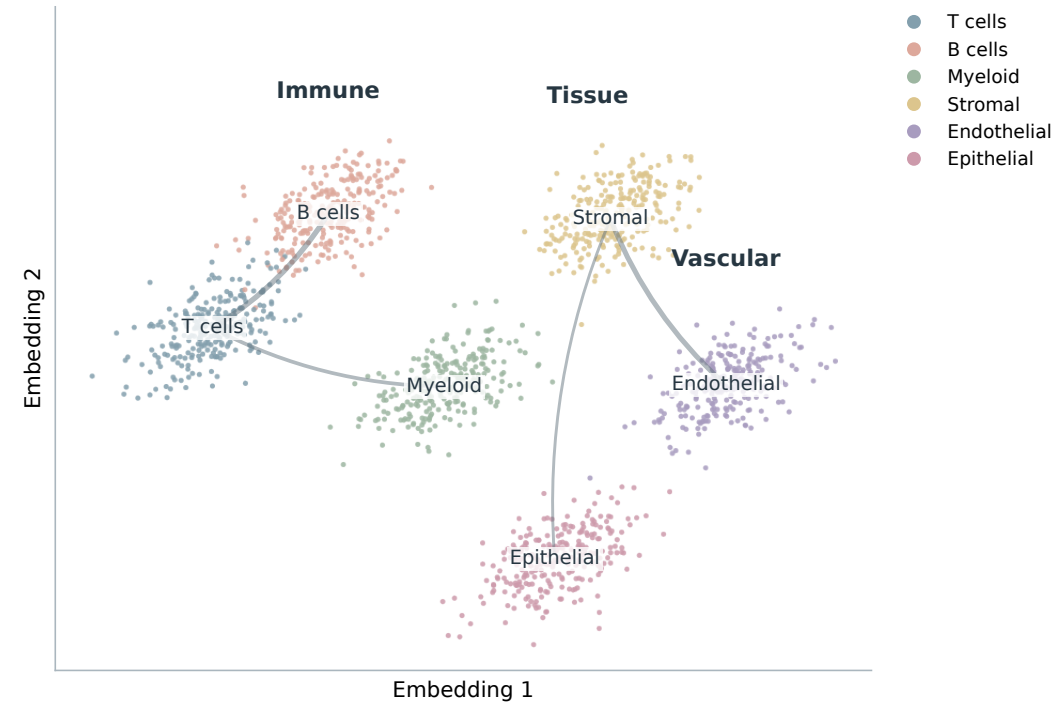


Cell identity and lineage hierarchy



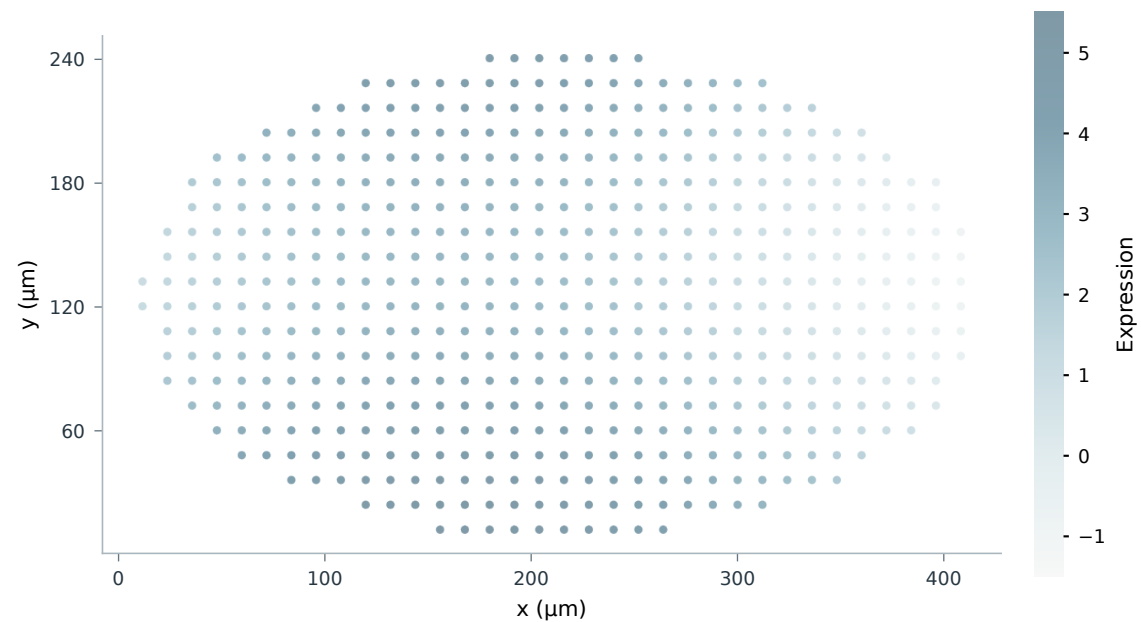
SYNTHETIC EXAMPLE

Cell identity and lineage hierarchy



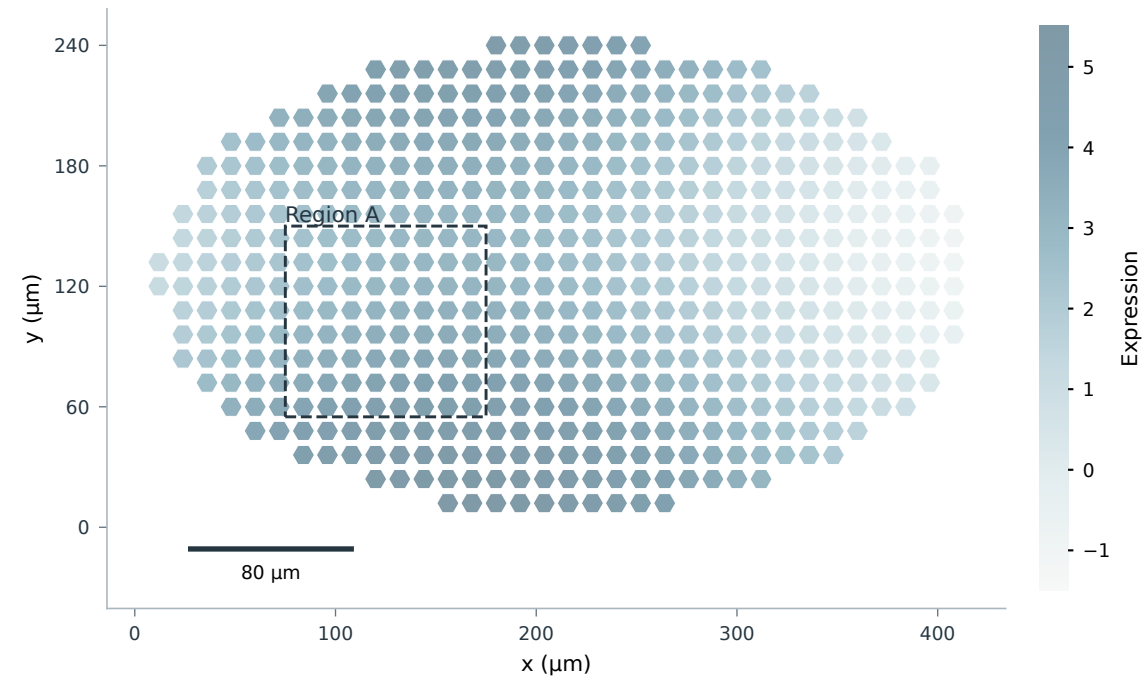
SYNTHETIC EXAMPLE

Spatial expression and tissue compartments



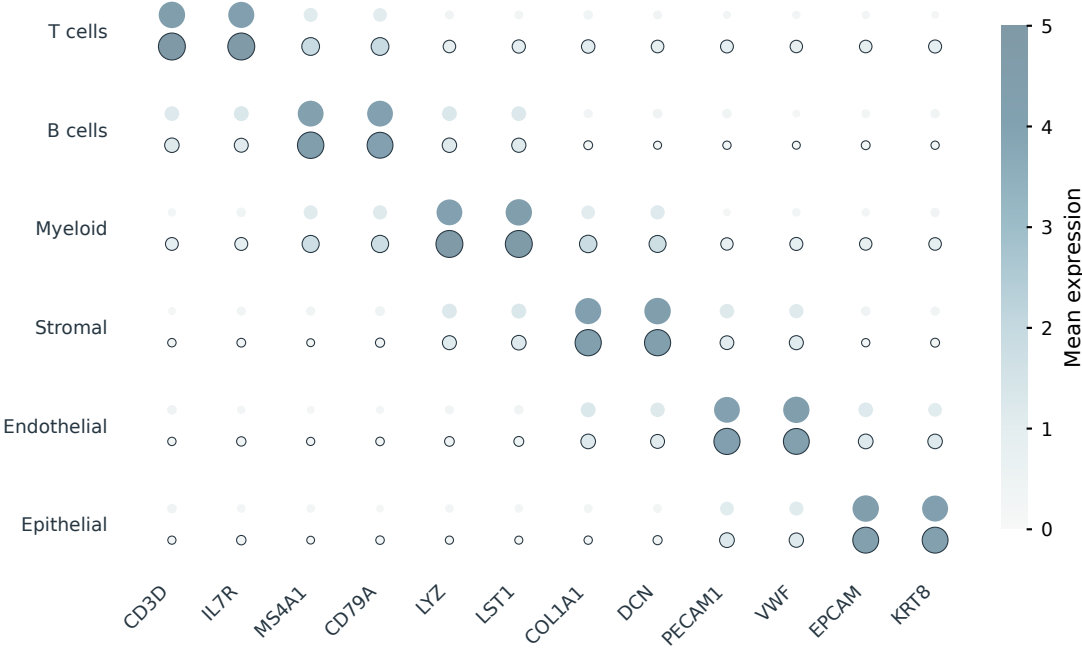
SYNTHETIC EXAMPLE

Spatial expression and tissue compartments



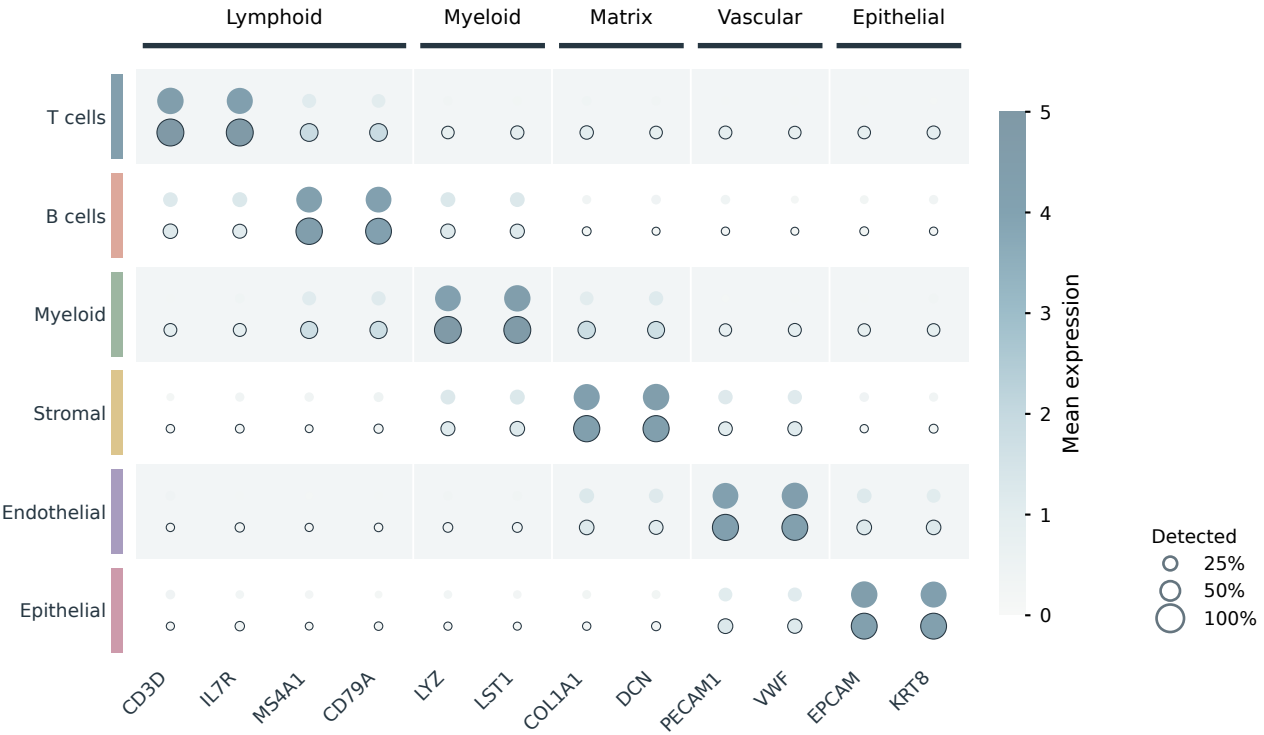
SYNTHETIC EXAMPLE

Marker programmes across cell states



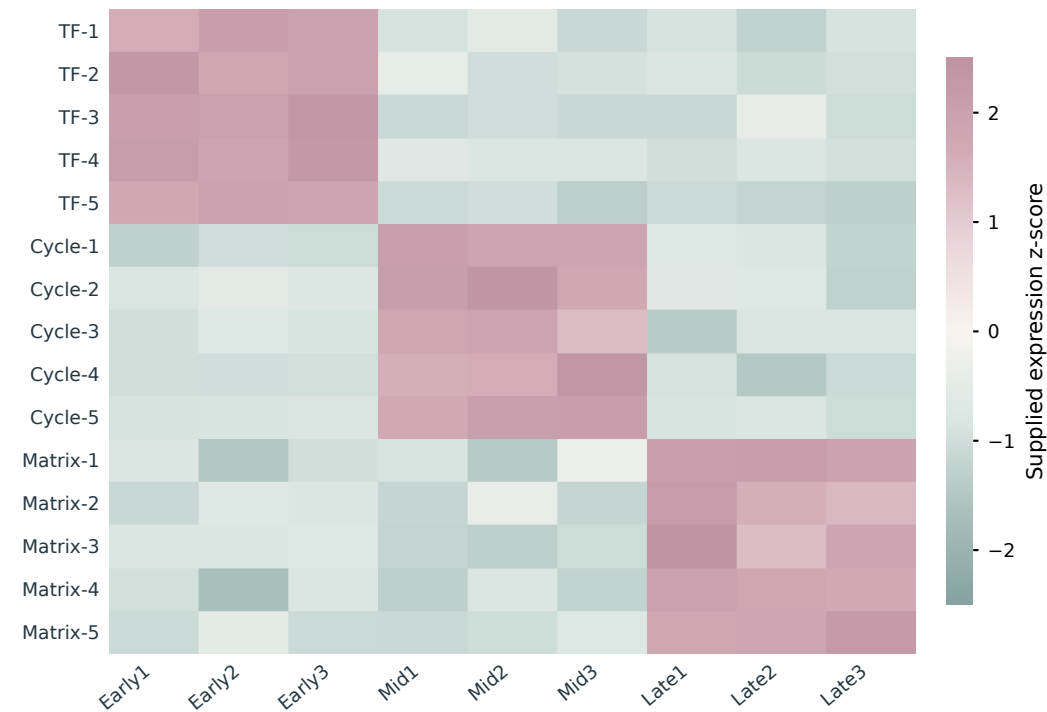
SYNTHETIC EXAMPLE

Marker programmes across cell states



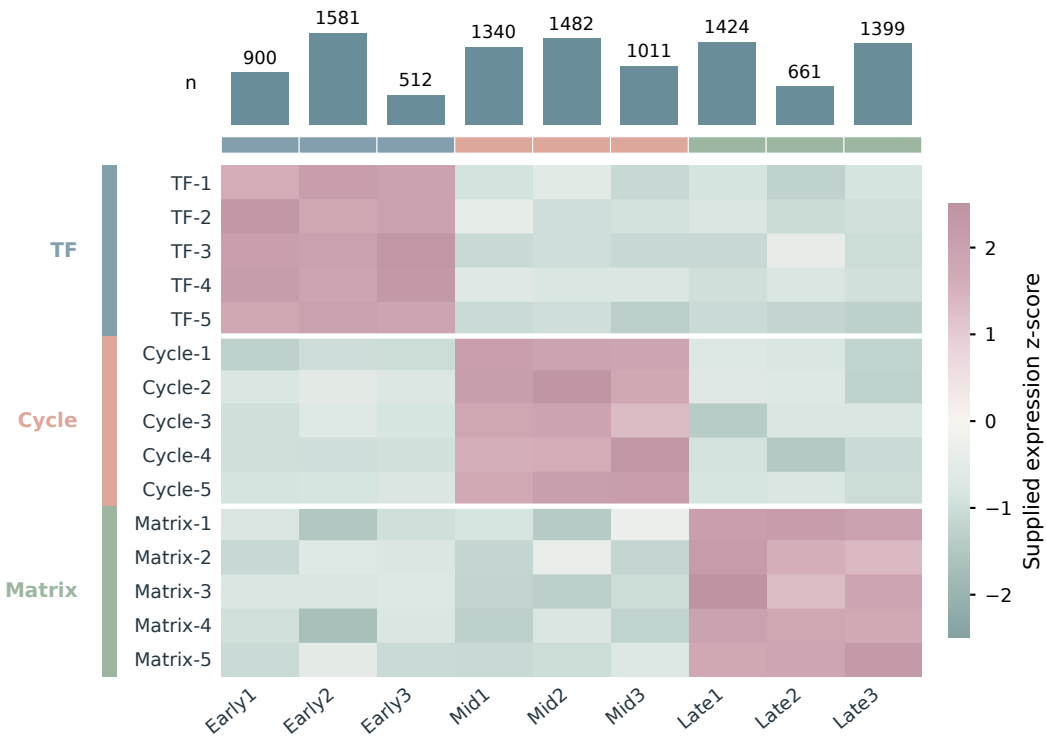
SYNTHETIC EXAMPLE

Expression modules and sample context



SYNTHETIC EXAMPLE

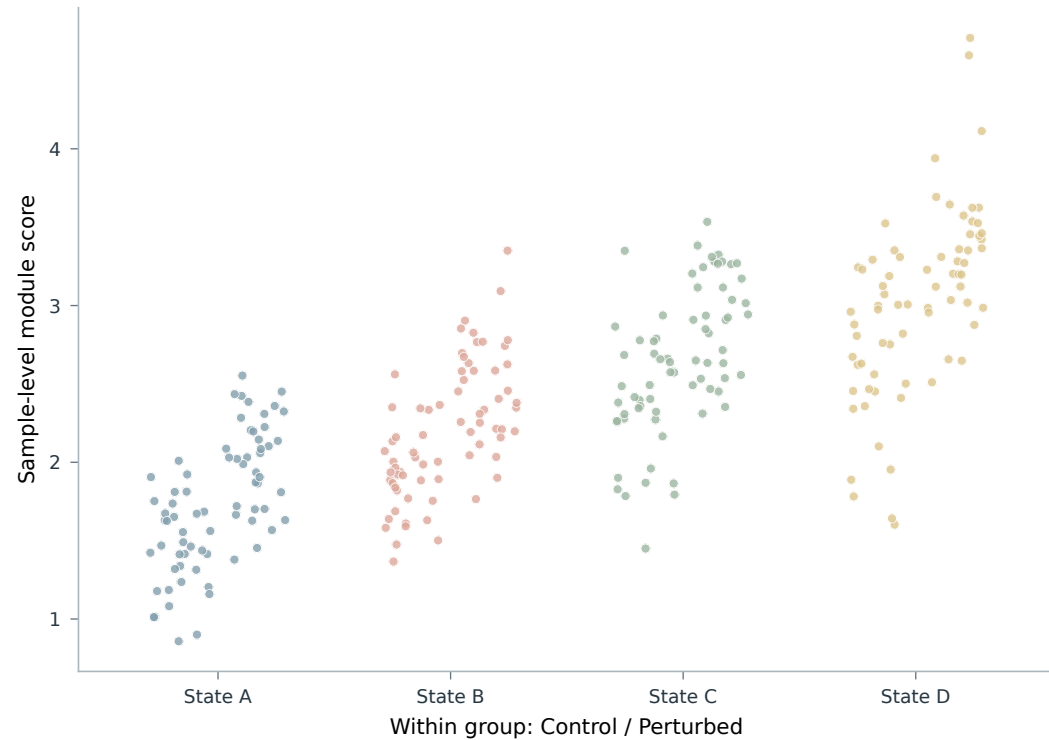
Expression modules and sample context



SYNTHETIC EXAMPLE

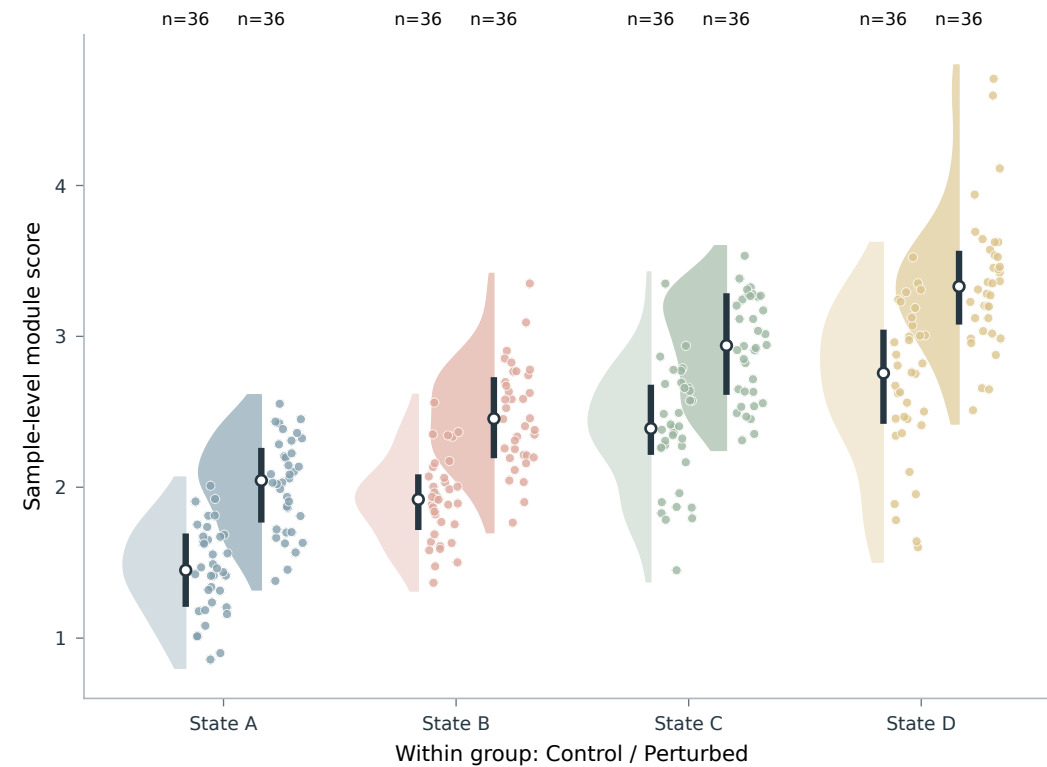
Condition
Early Mid Late

Distribution across biological samples



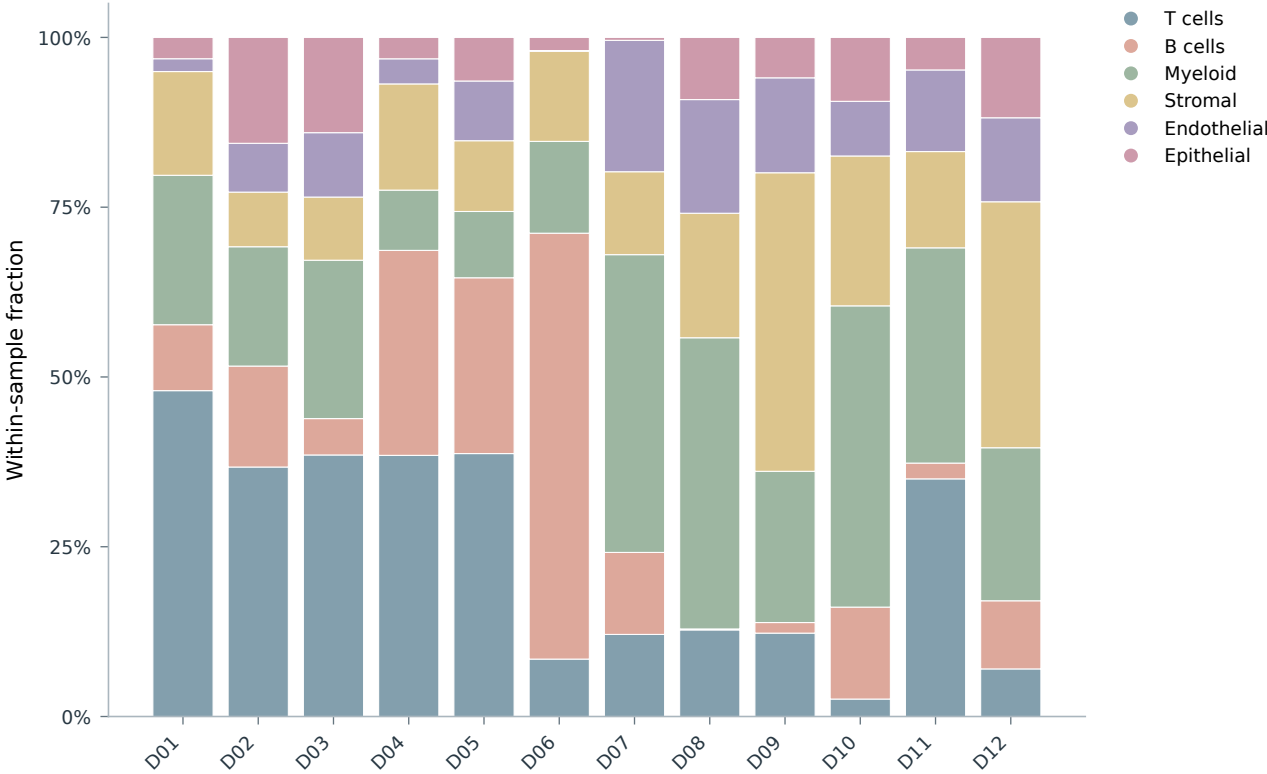
SYNTHETIC EXAMPLE

Distribution across biological samples



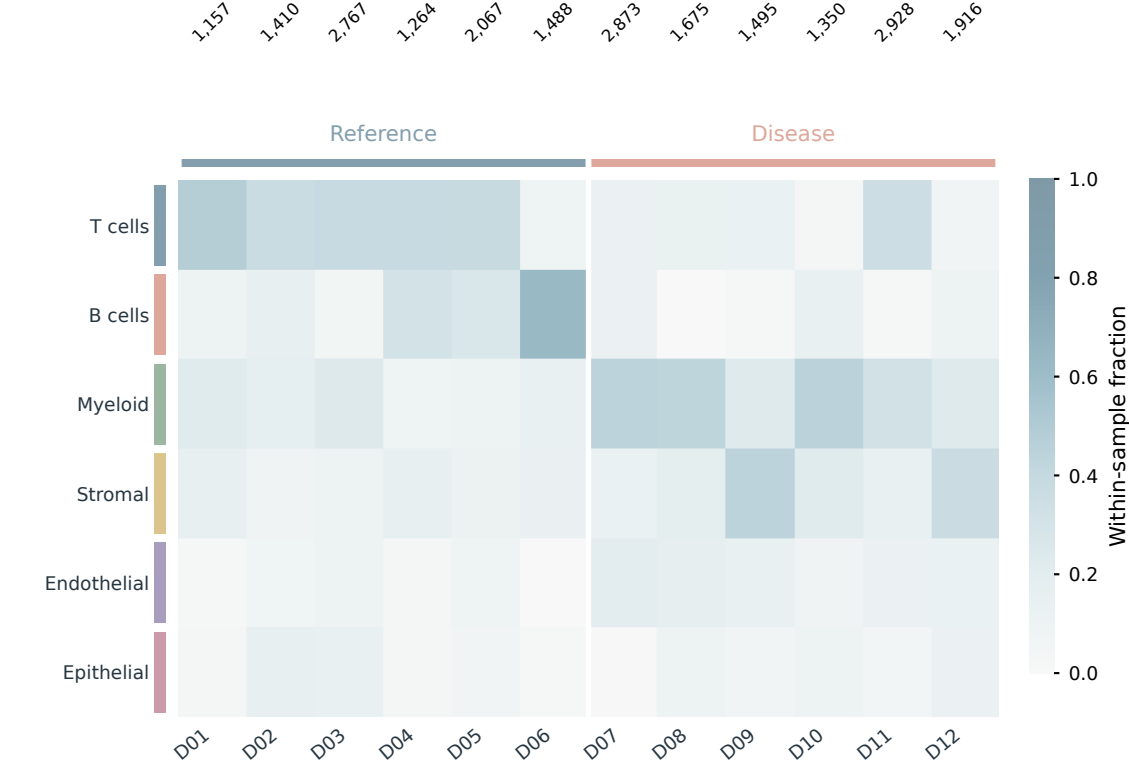
SYNTHETIC EXAMPLE

Cell composition across samples



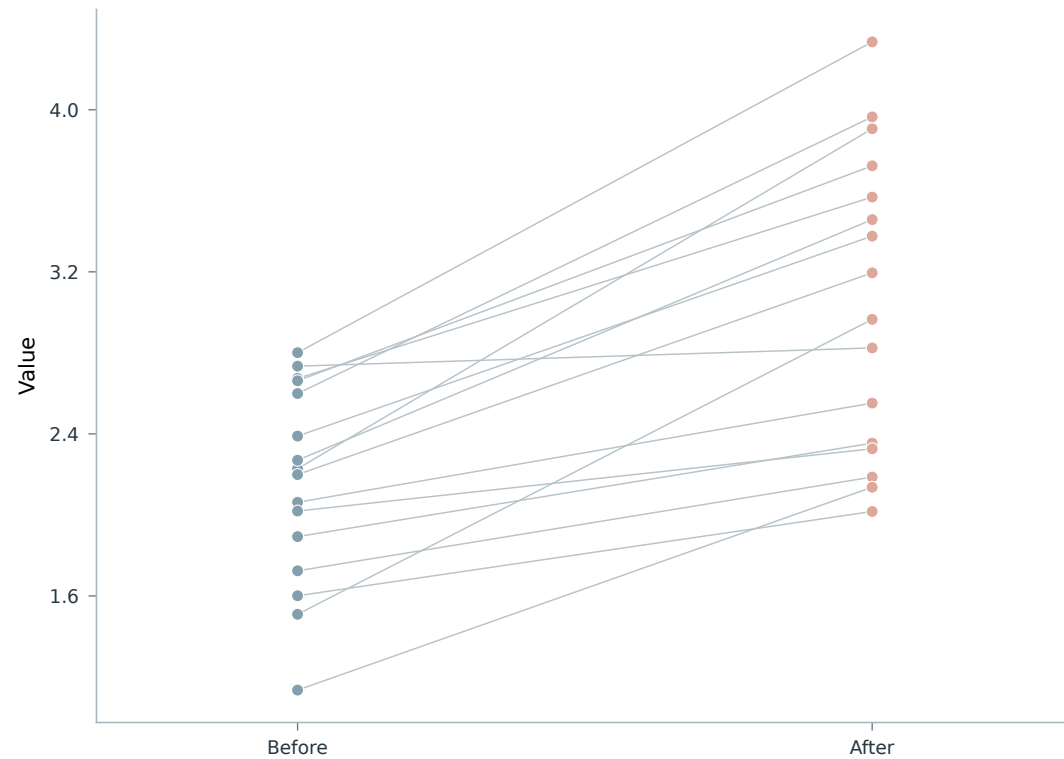
SYNTHETIC EXAMPLE

Cell composition across samples



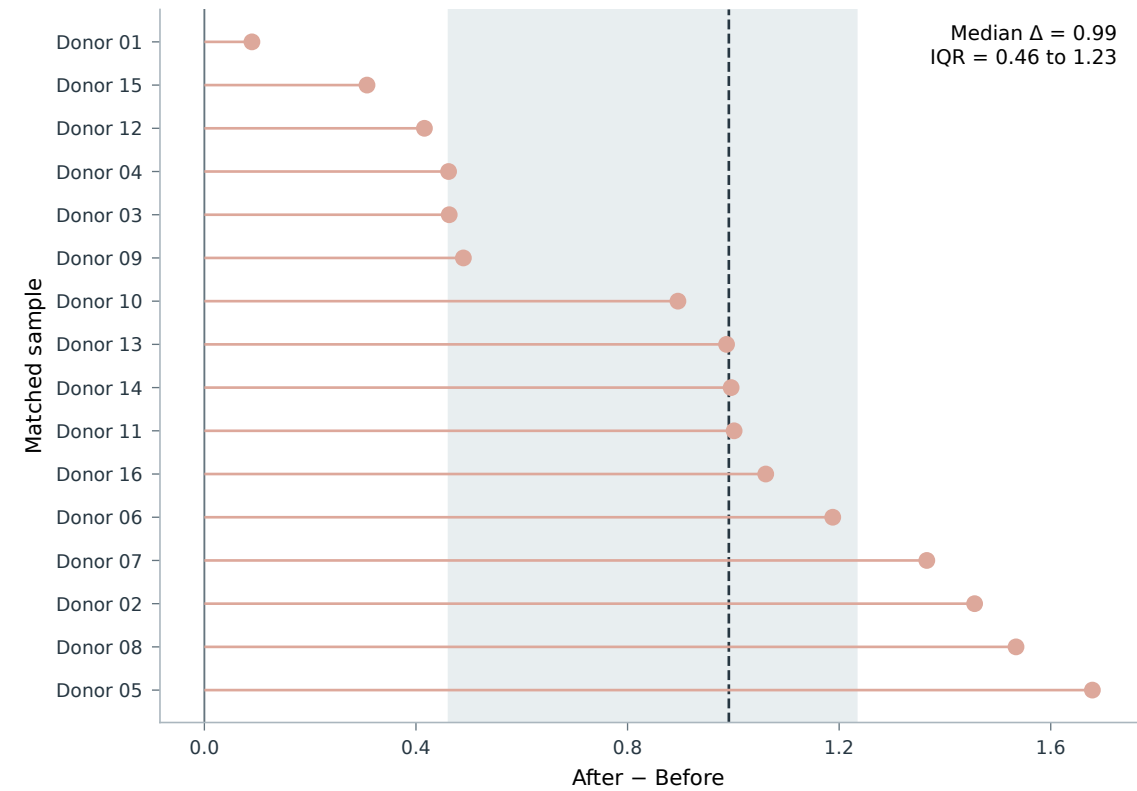
SYNTHETIC EXAMPLE

Matched sample responses



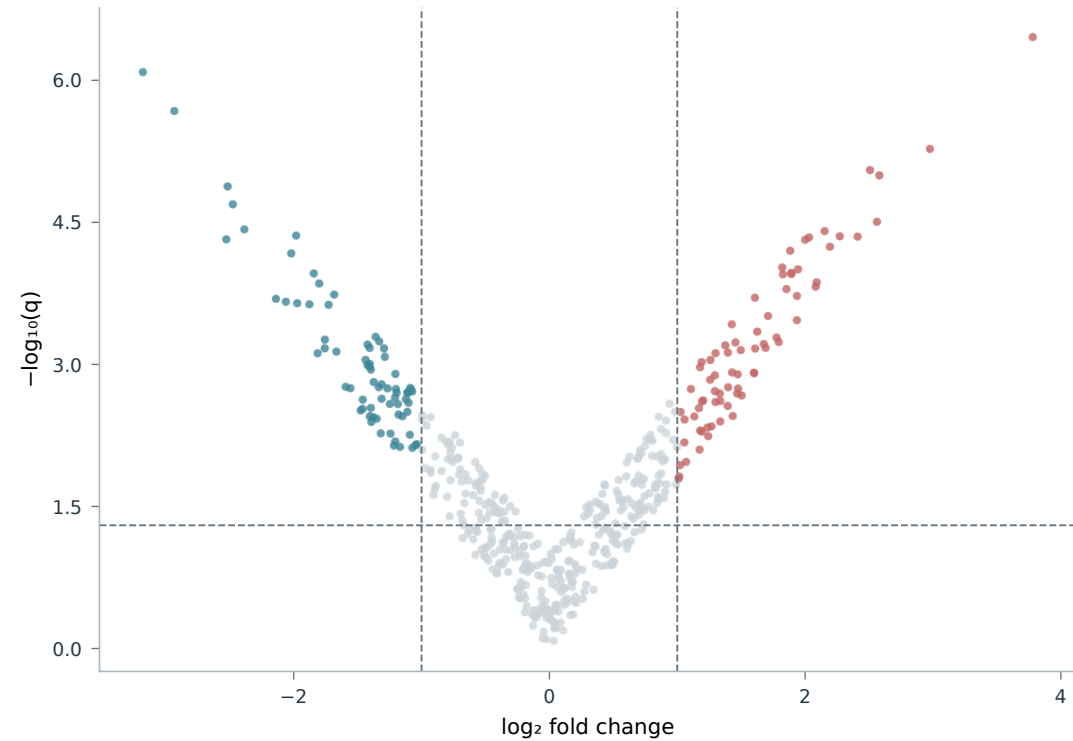
SYNTHETIC EXAMPLE

Matched sample responses



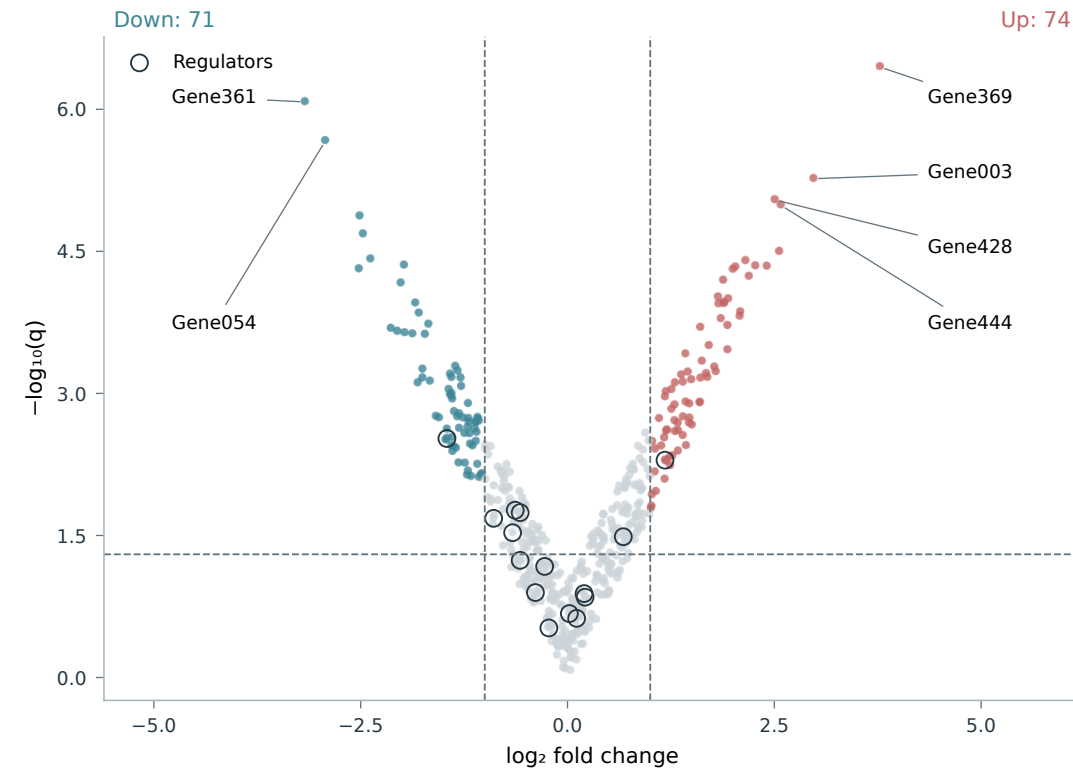
SYNTHETIC EXAMPLE

Differential expression and annotated gene sets



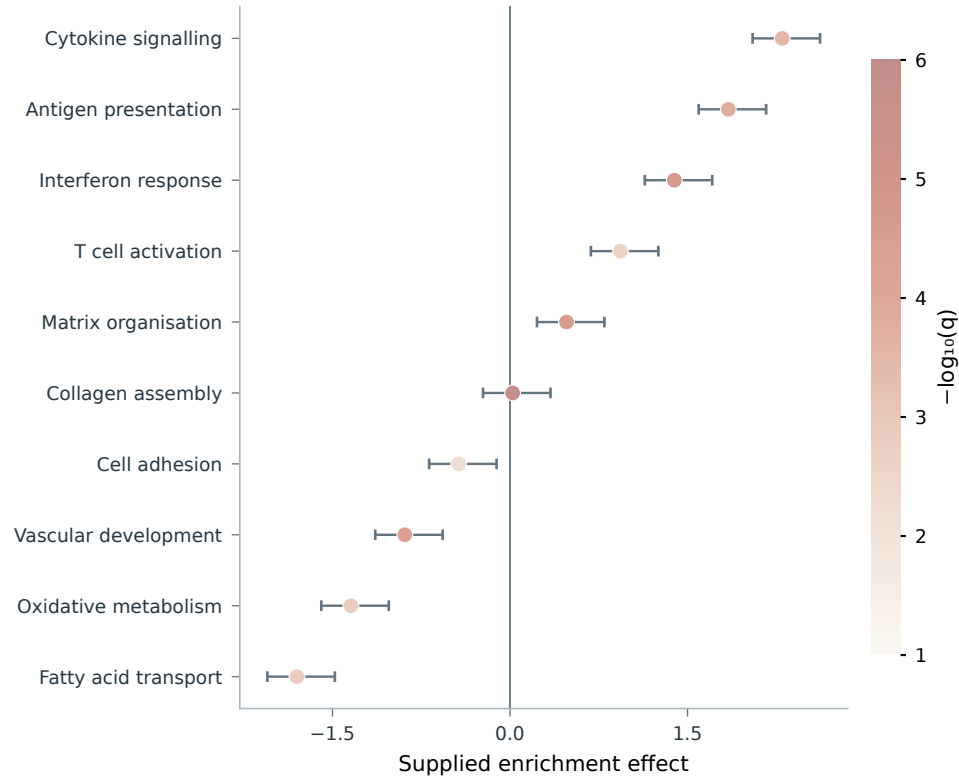
SYNTHETIC EXAMPLE

Differential expression and annotated gene sets



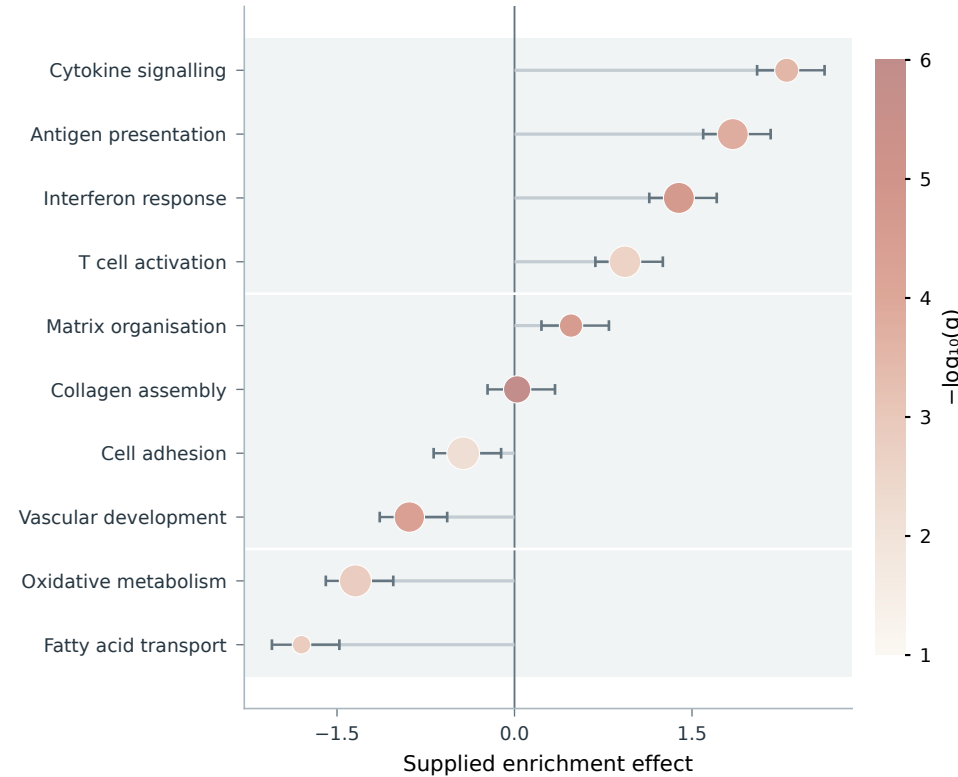
SYNTHETIC EXAMPLE

Pathway effects and supporting evidence



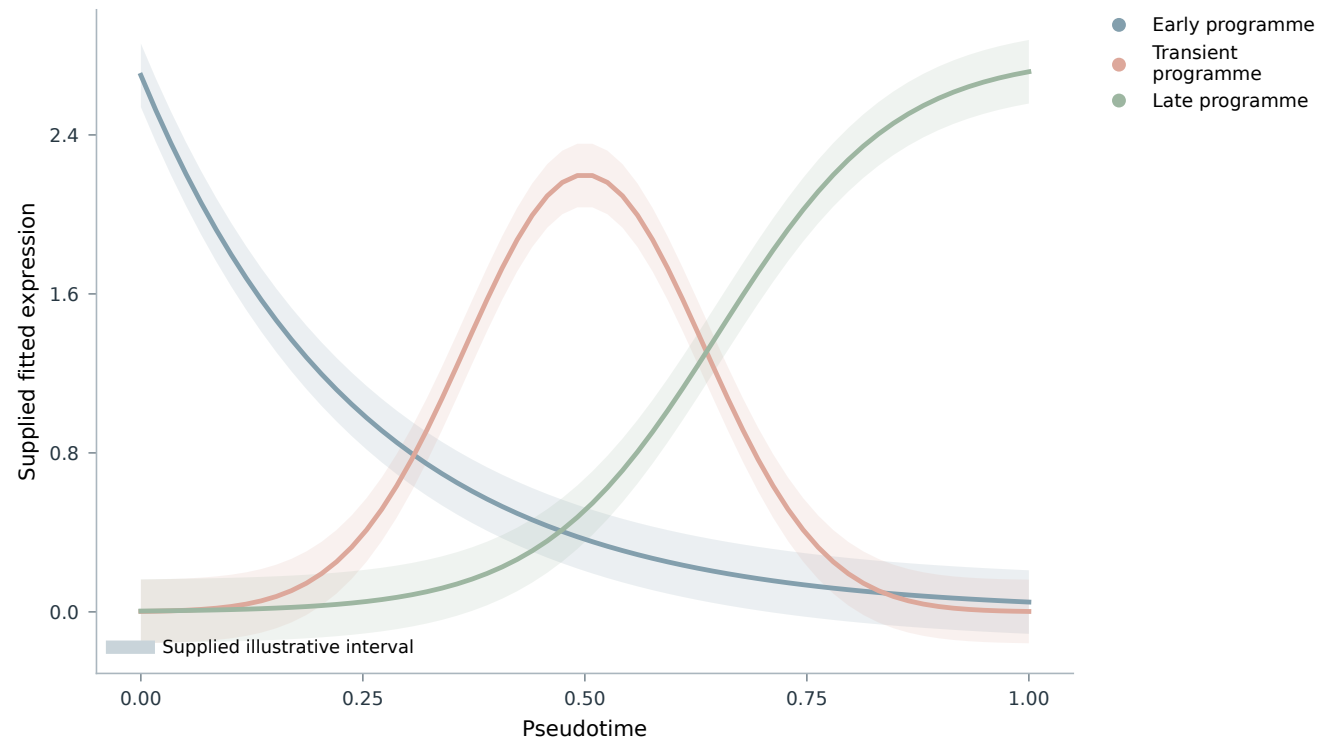
SYNTHETIC EXAMPLE

Pathway effects and supporting evidence



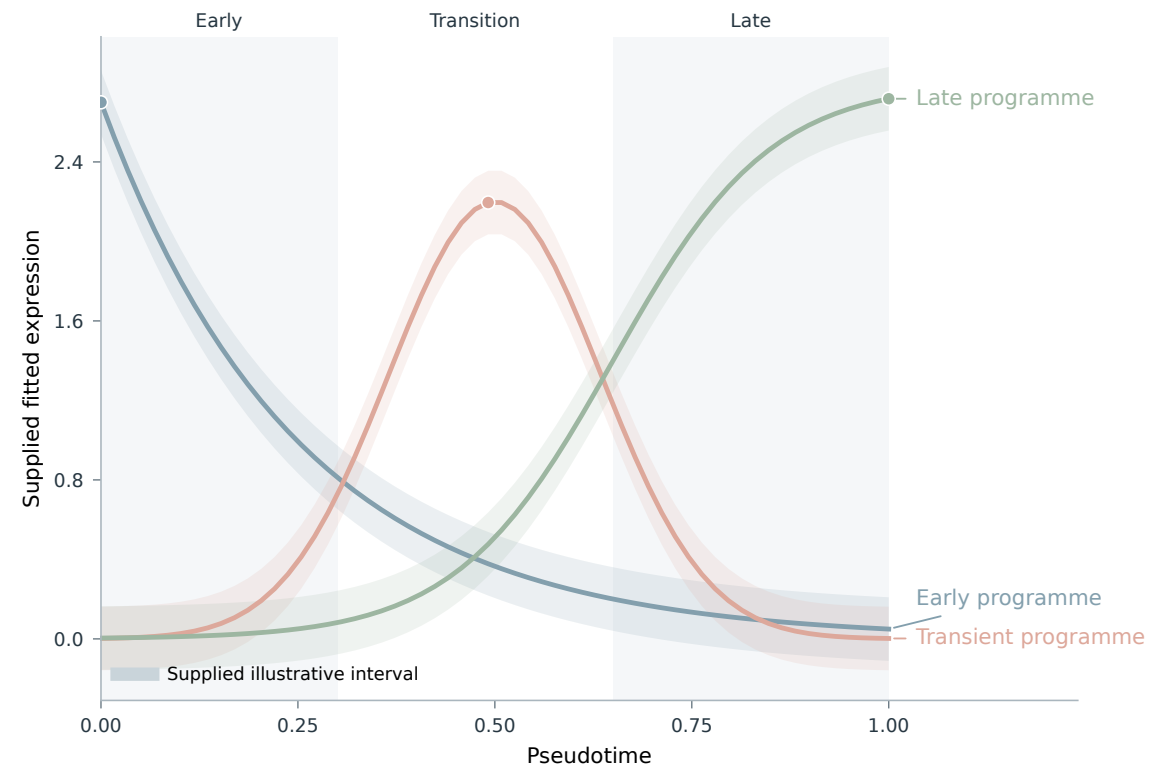
SYNTHETIC EXAMPLE

Lineage programmes over time



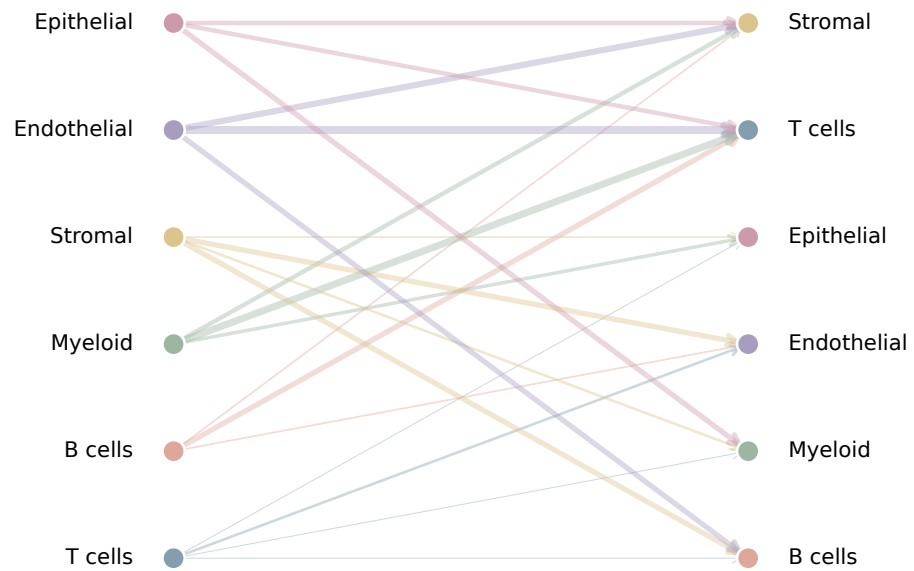
SYNTHETIC EXAMPLE

Lineage programmes over time



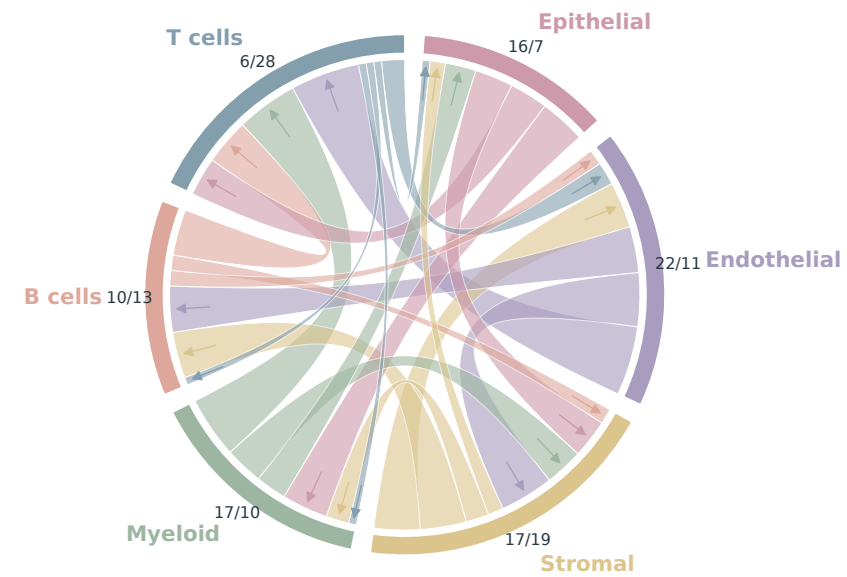
SYNTHETIC EXAMPLE

Candidate sender-receiver communication



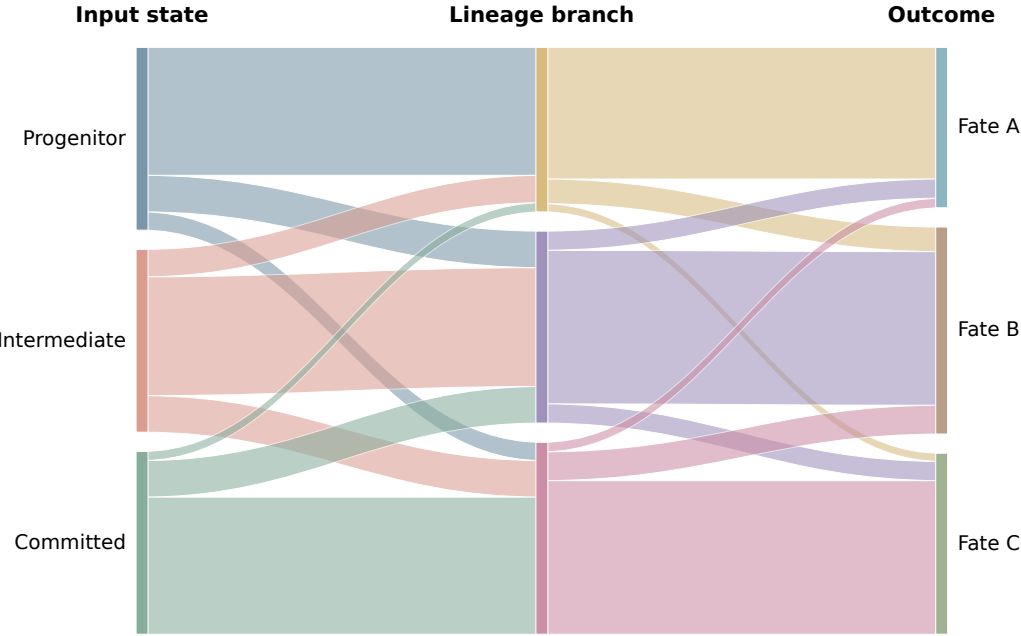
SYNTHETIC EXAMPLE

Candidate sender-receiver communication



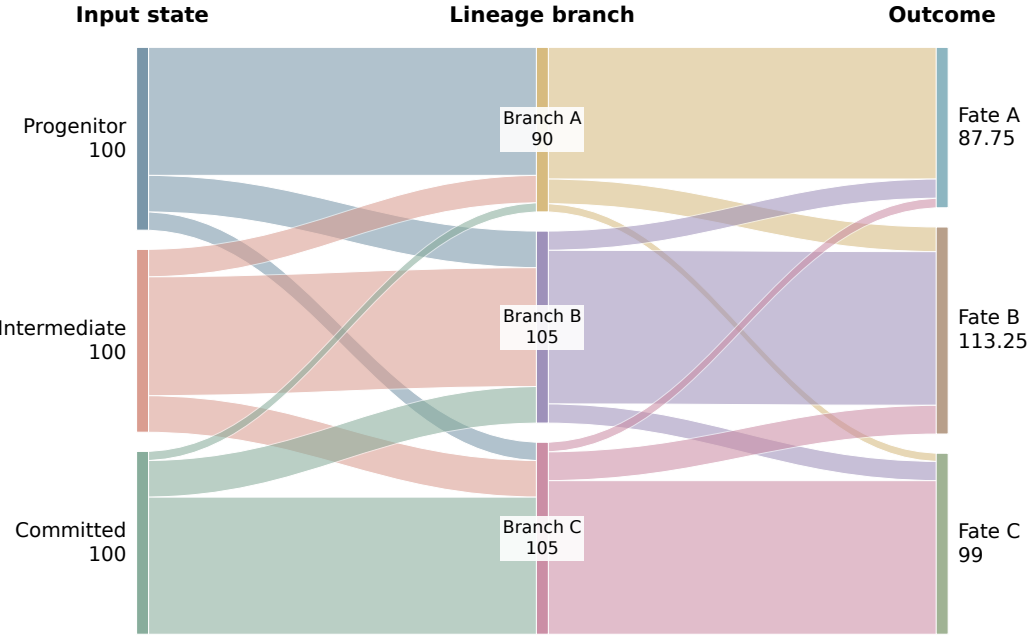
SYNTHETIC EXAMPLE

State transitions and conserved mass



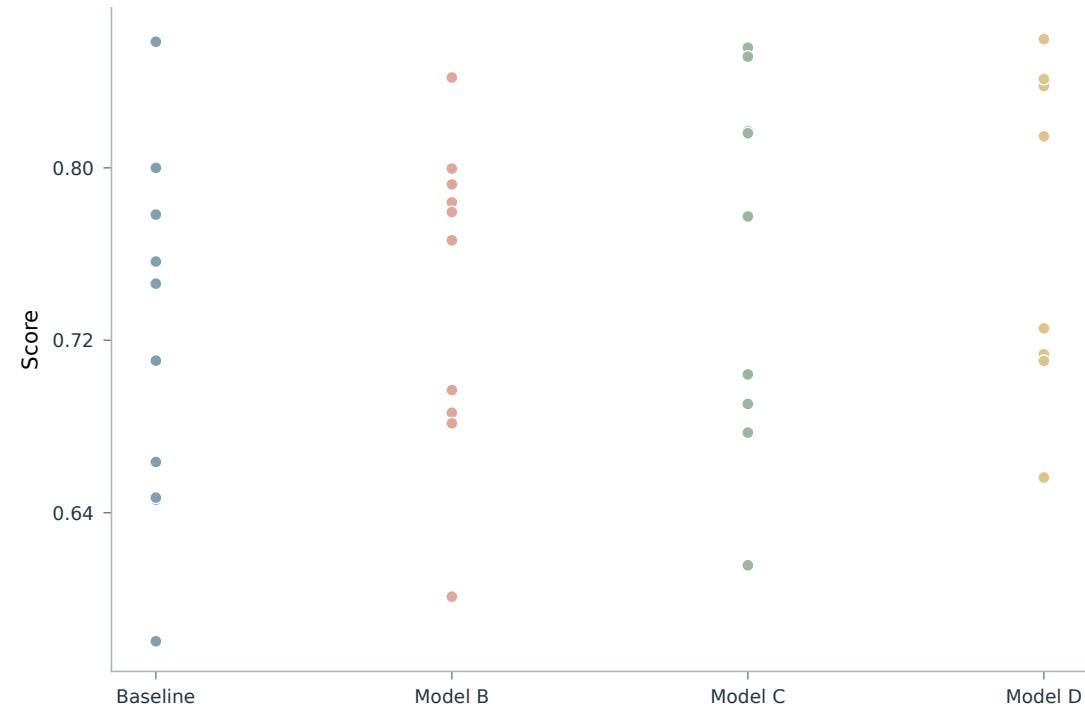
SYNTHETIC EXAMPLE

State transitions and conserved mass



SYNTHETIC EXAMPLE

Matched-dataset method performance



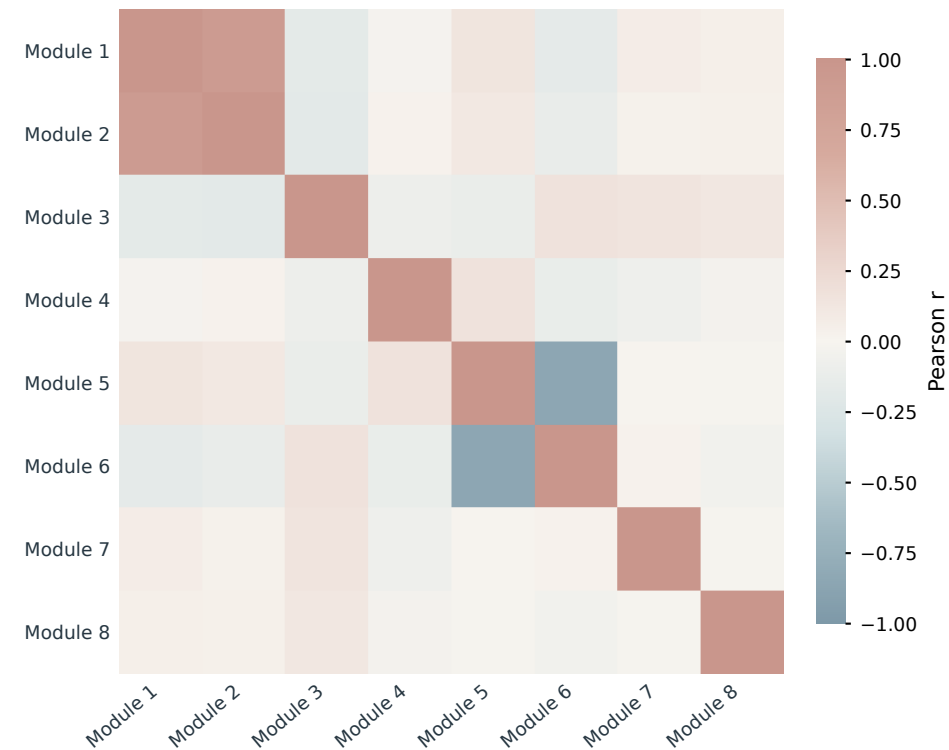
SYNTHETIC EXAMPLE

Matched-dataset method performance



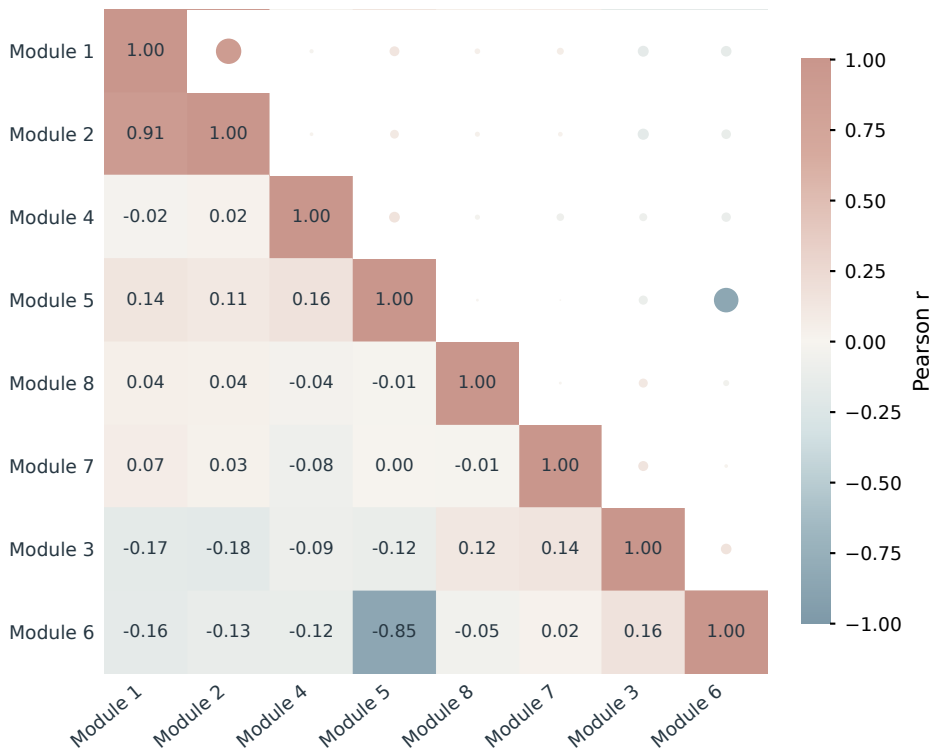
SYNTHETIC EXAMPLE

Correlation structure



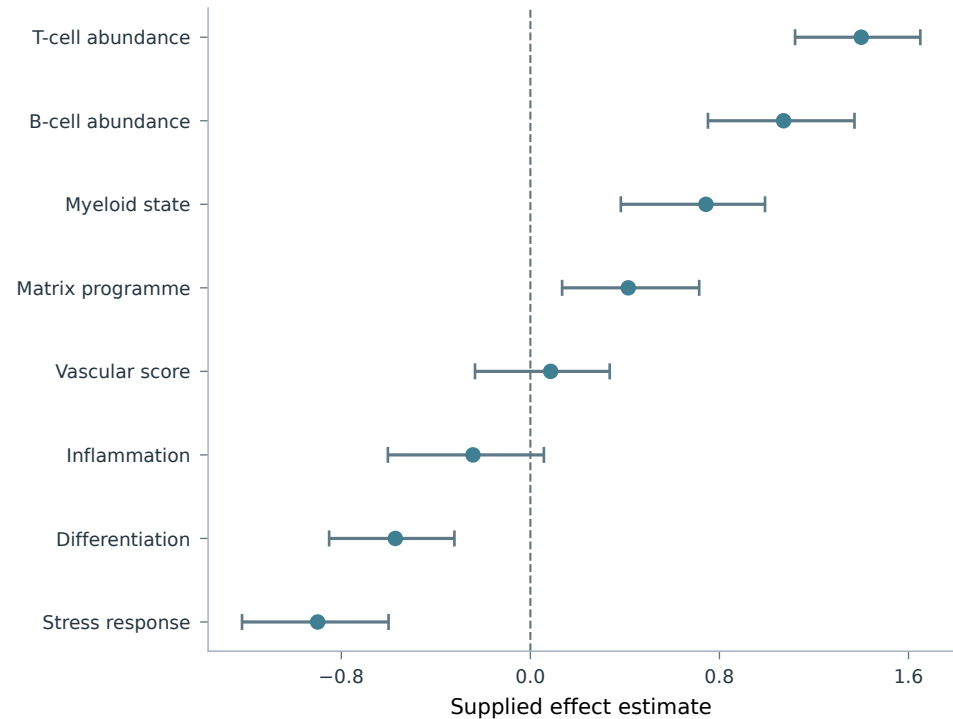
SYNTHETIC EXAMPLE

Correlation structure



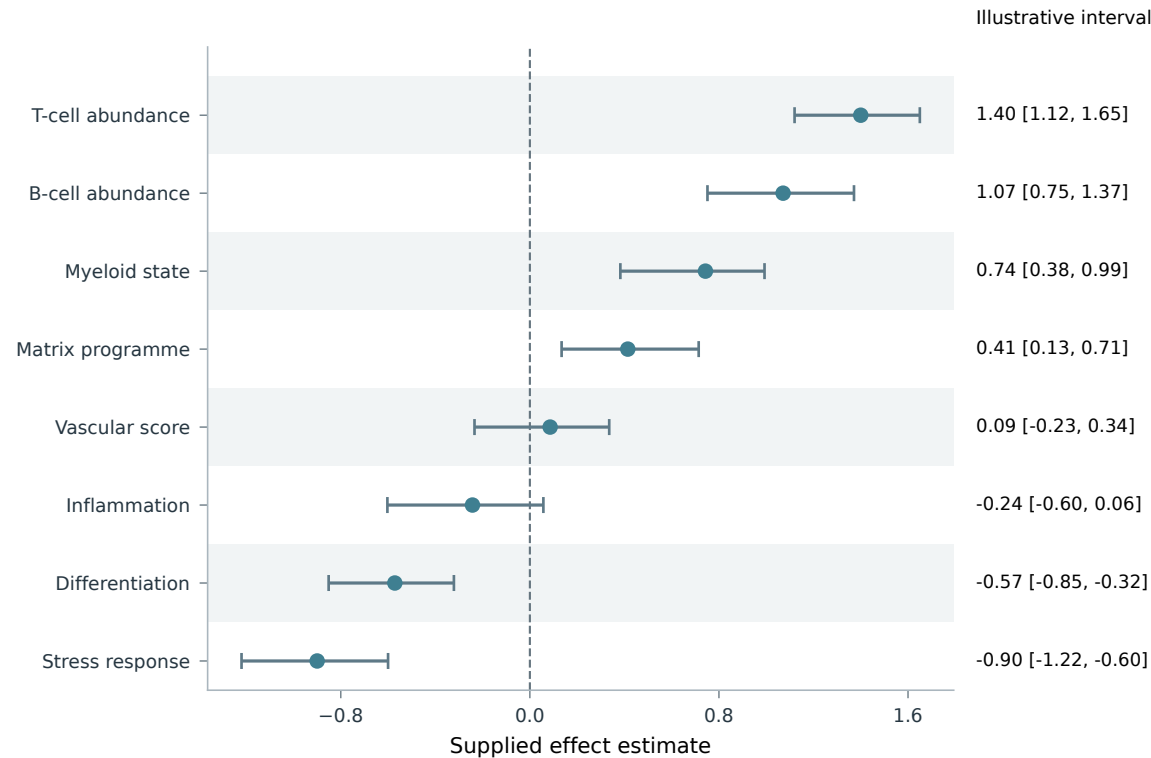
SYNTHETIC EXAMPLE

Effect sizes and uncertainty



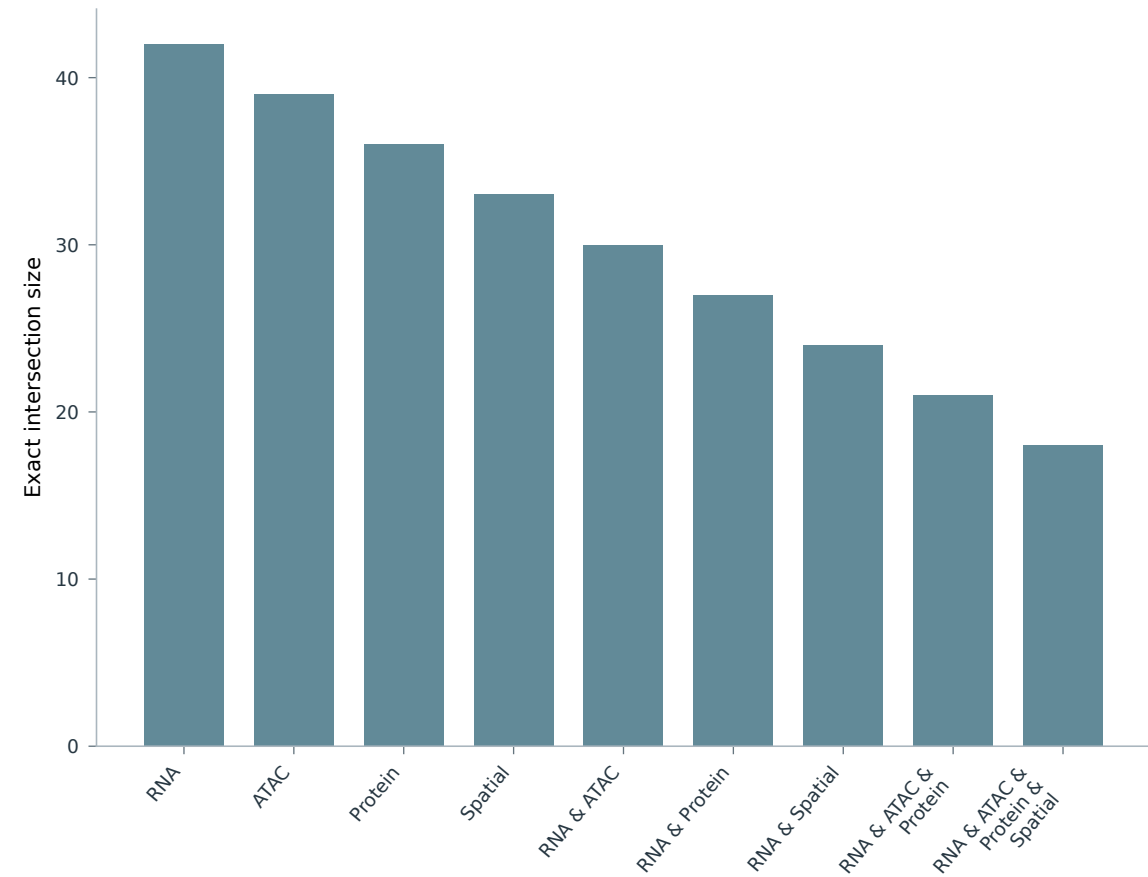
SYNTHETIC EXAMPLE

Effect sizes and uncertainty



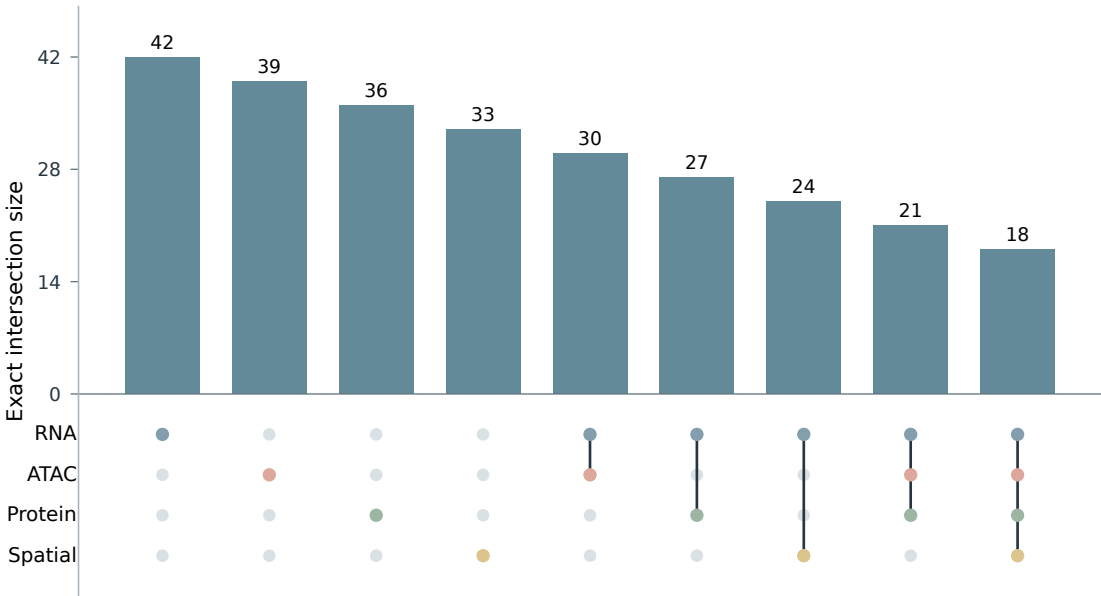
SYNTHETIC EXAMPLE

Exact intersections between gene sets



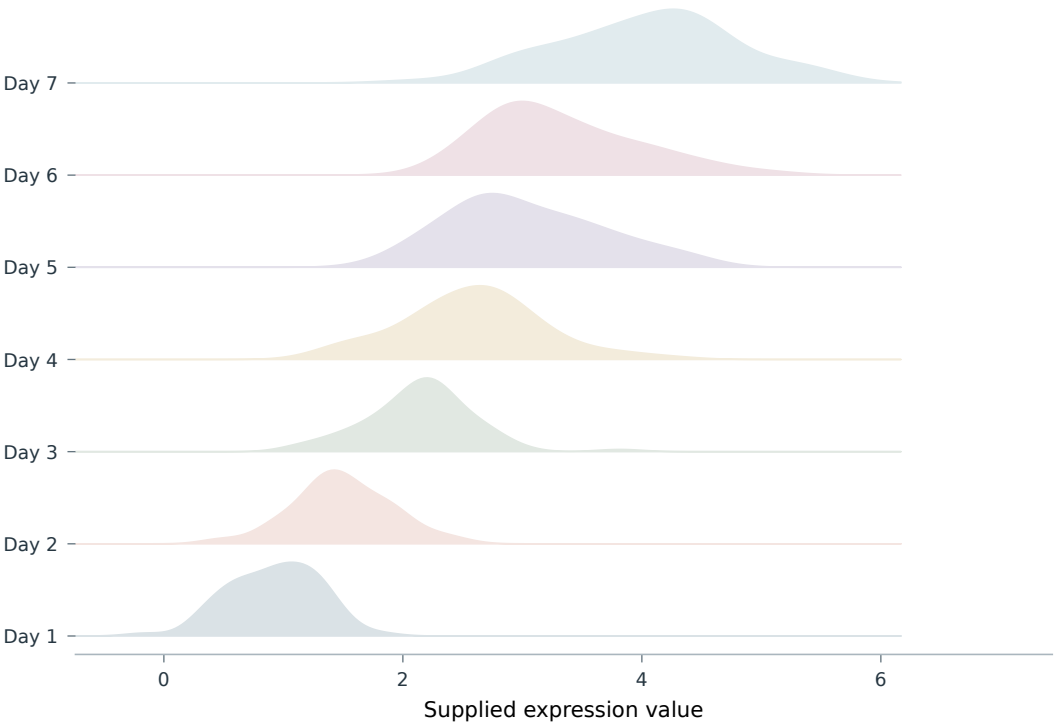
SYNTHETIC EXAMPLE

Exact intersections between gene sets



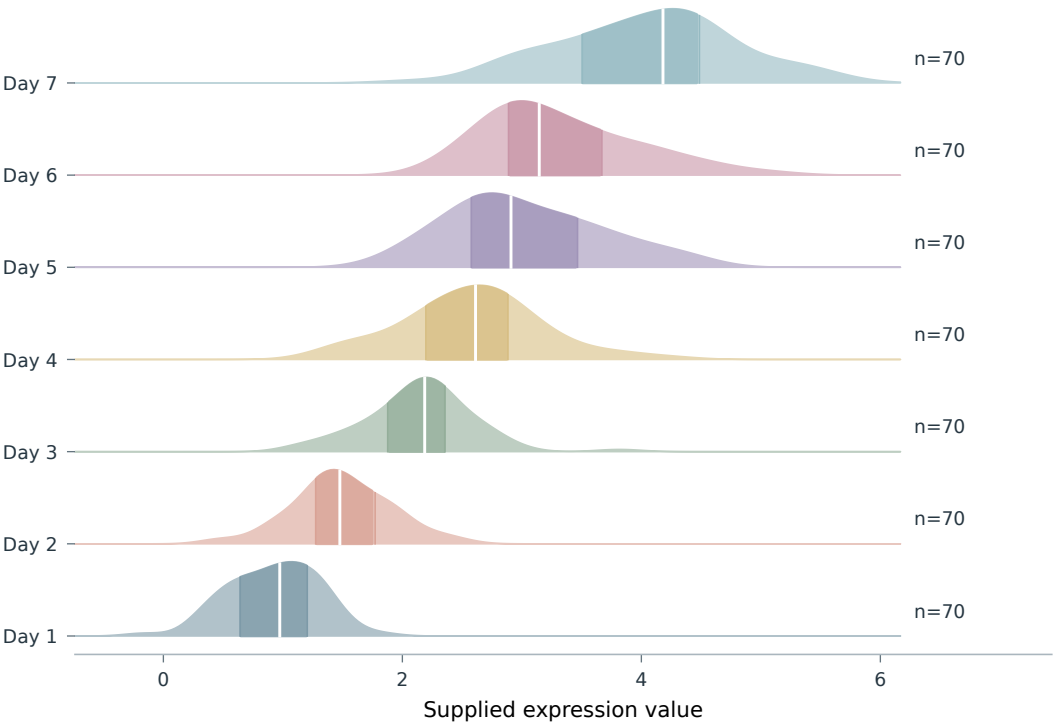
SYNTHETIC EXAMPLE

Expression distributions across ordered states



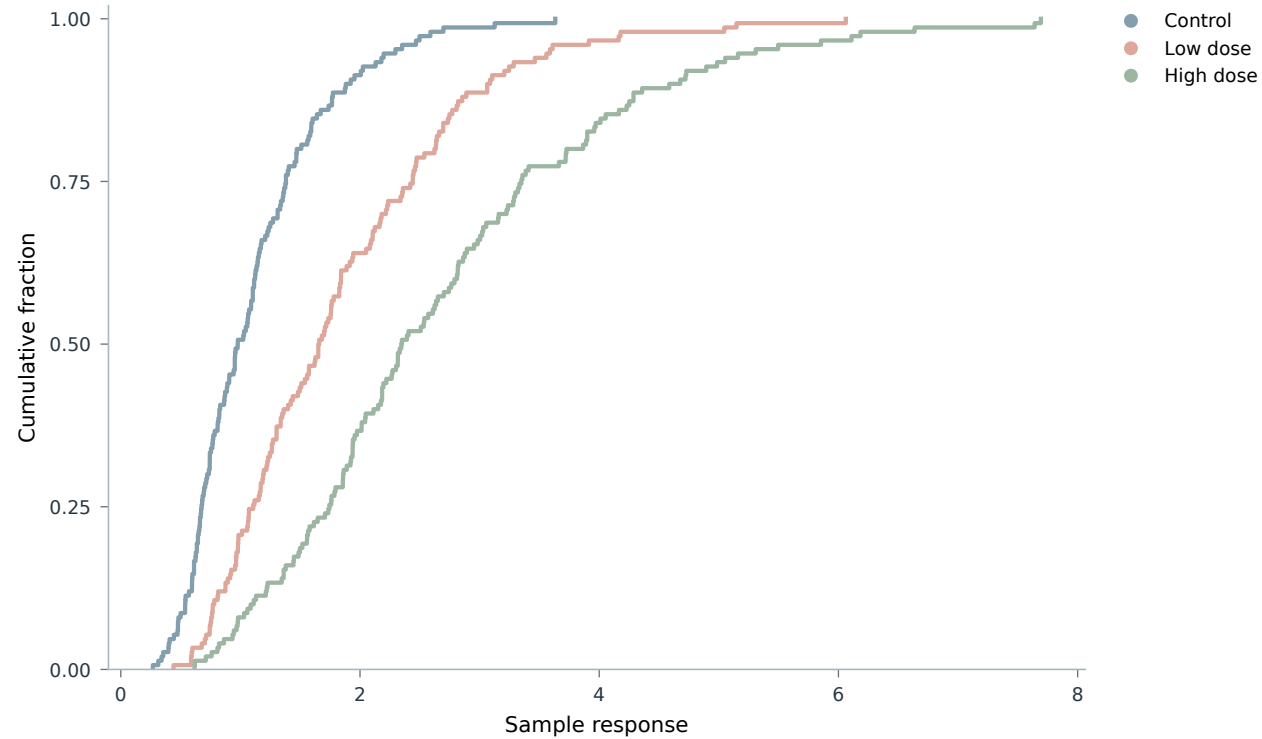
SYNTHETIC EXAMPLE

Expression distributions across ordered states



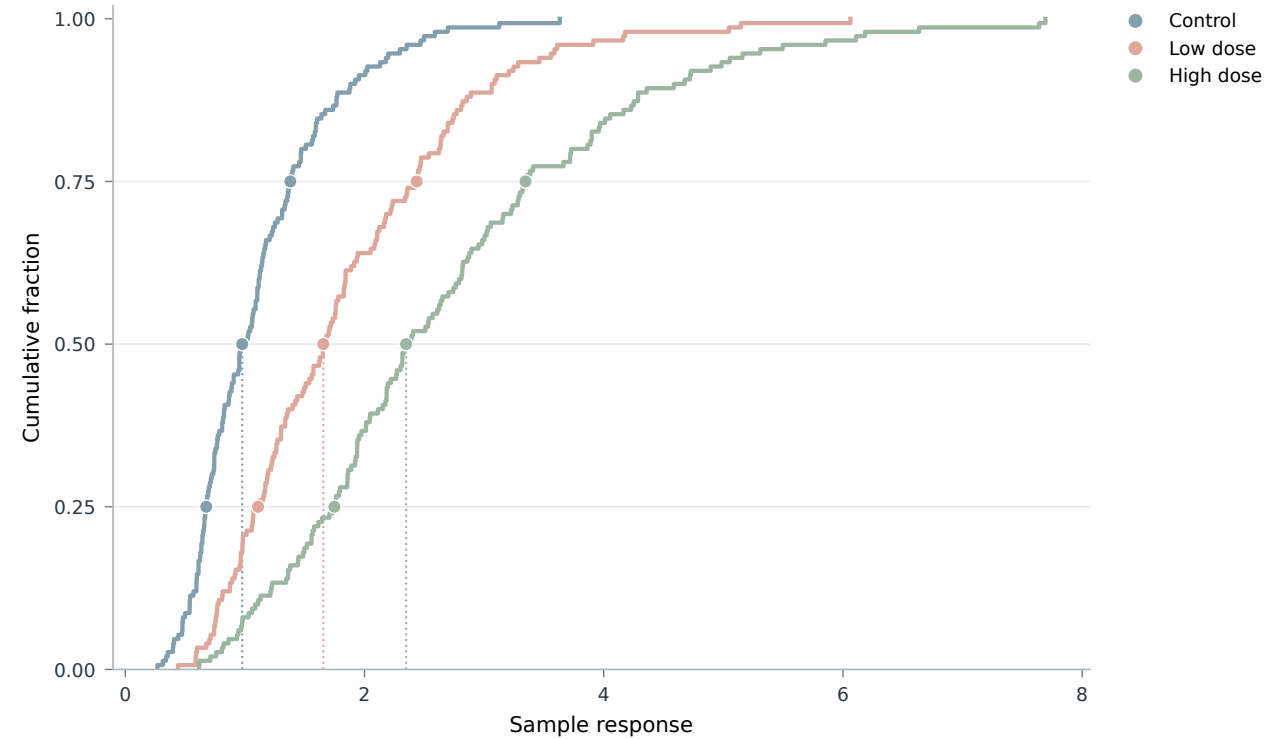
SYNTHETIC EXAMPLE

Cumulative sample distributions



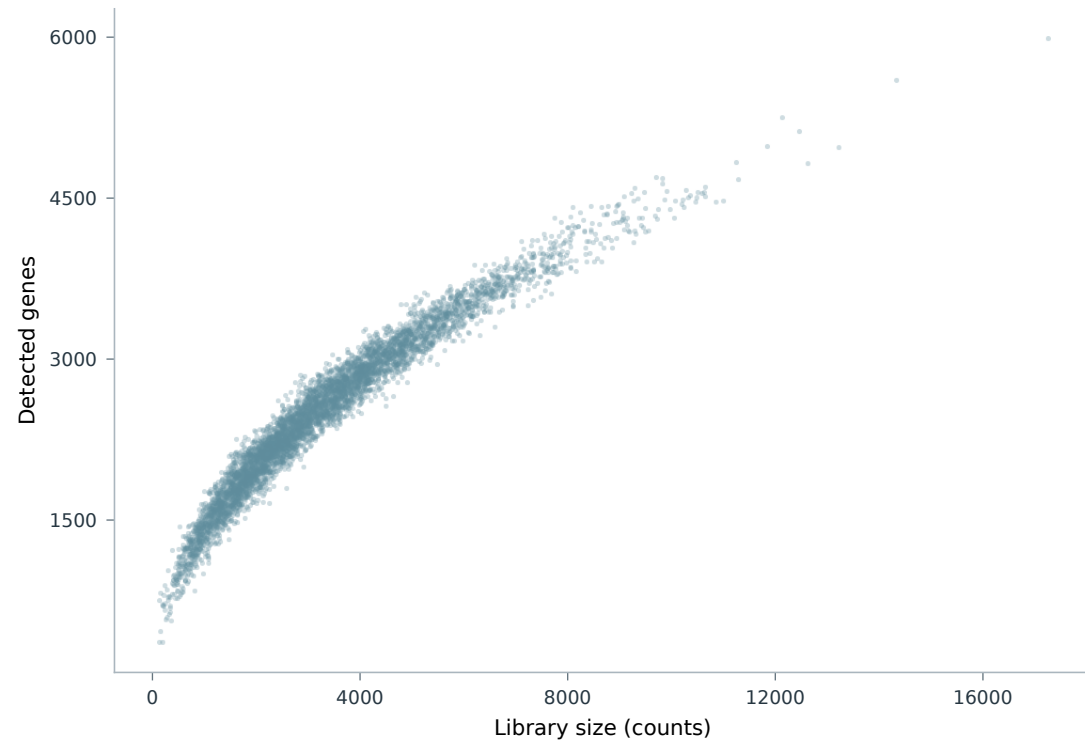
SYNTHETIC EXAMPLE

Cumulative sample distributions



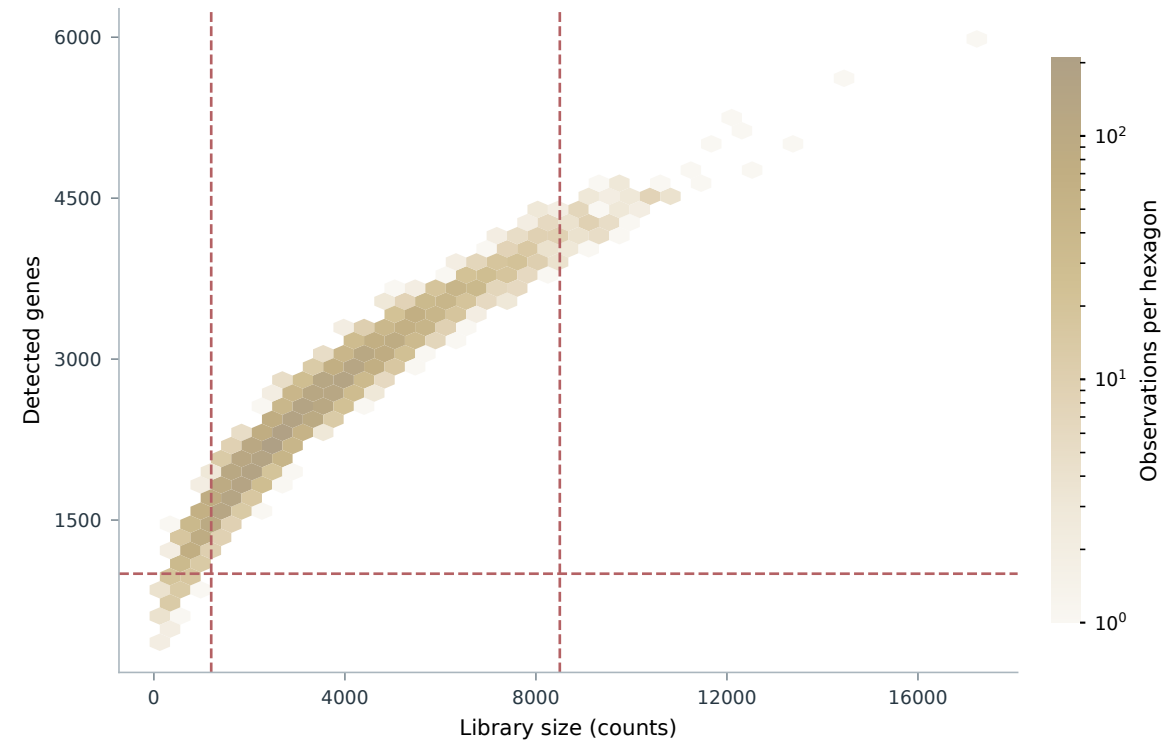
SYNTHETIC EXAMPLE

Density and quality-control thresholds



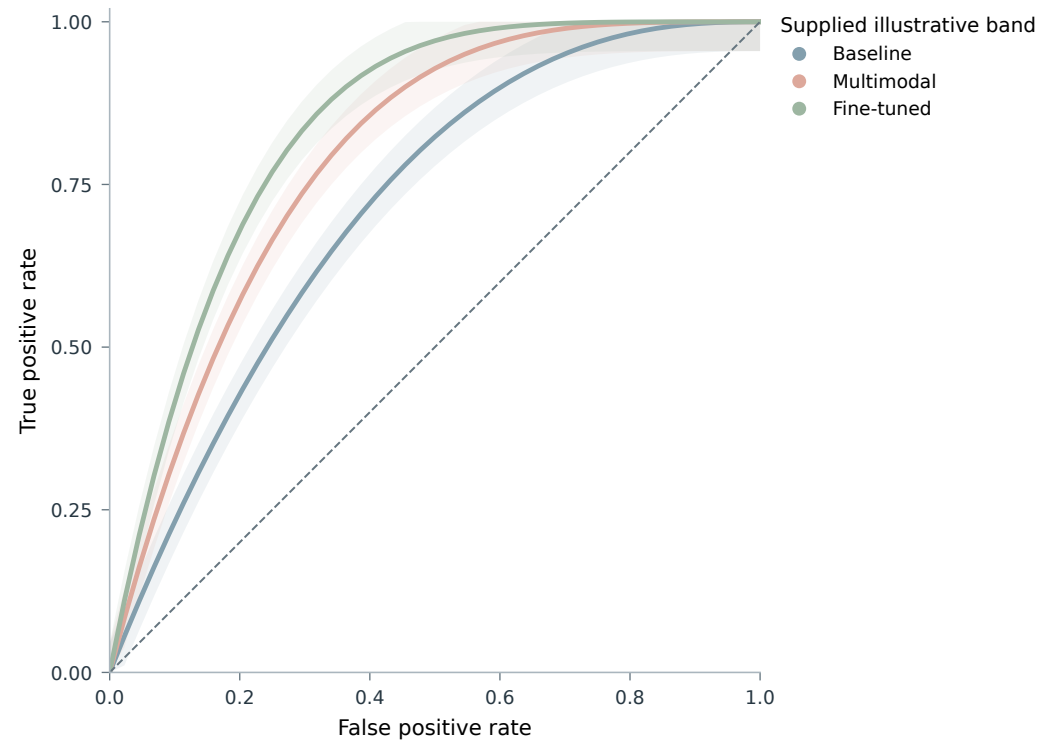
SYNTHETIC EXAMPLE

Density and quality-control thresholds



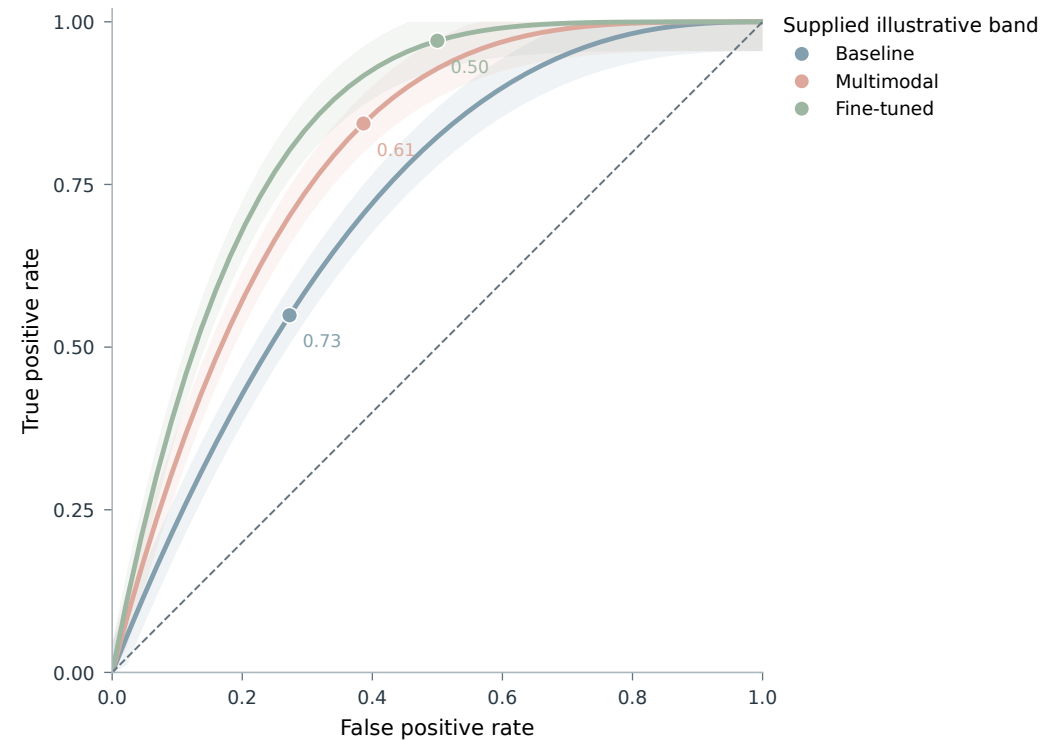
SYNTHETIC EXAMPLE

Discrimination across models



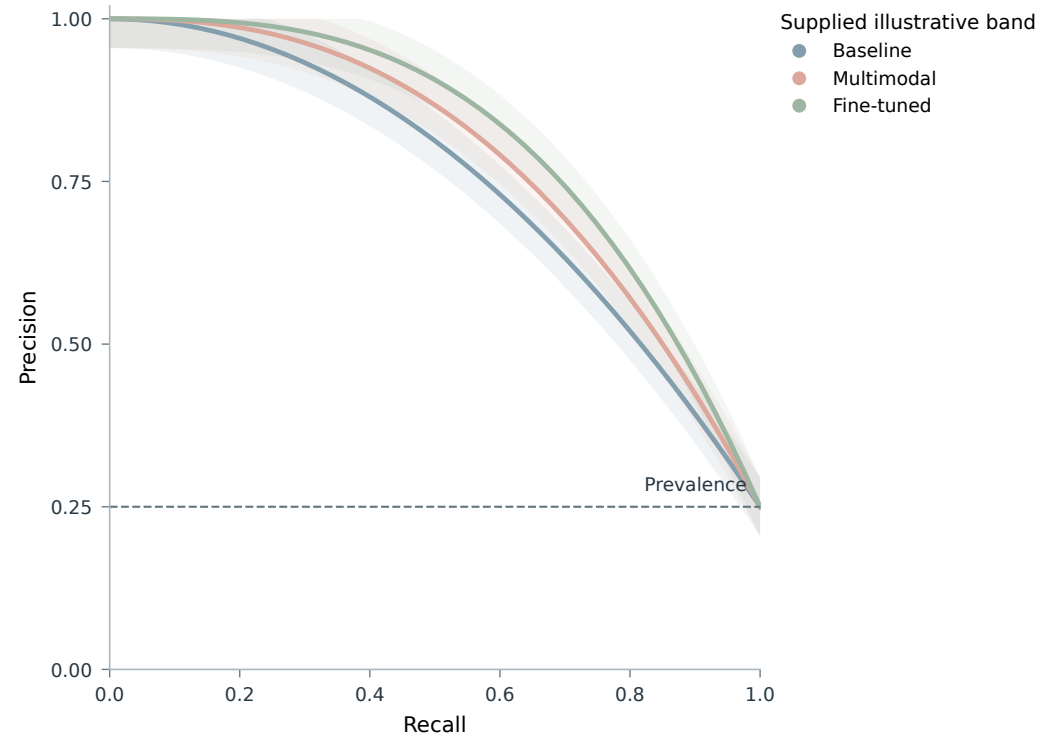
SYNTHETIC EXAMPLE

Discrimination across models



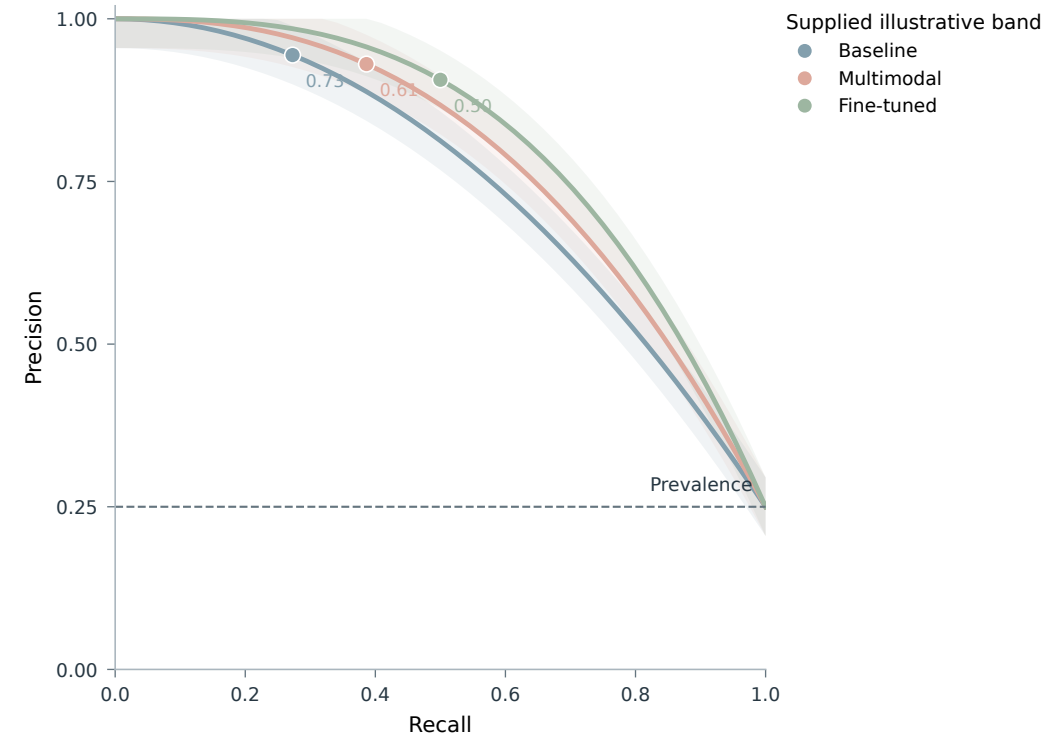
SYNTHETIC EXAMPLE

Precision-recall trade-offs



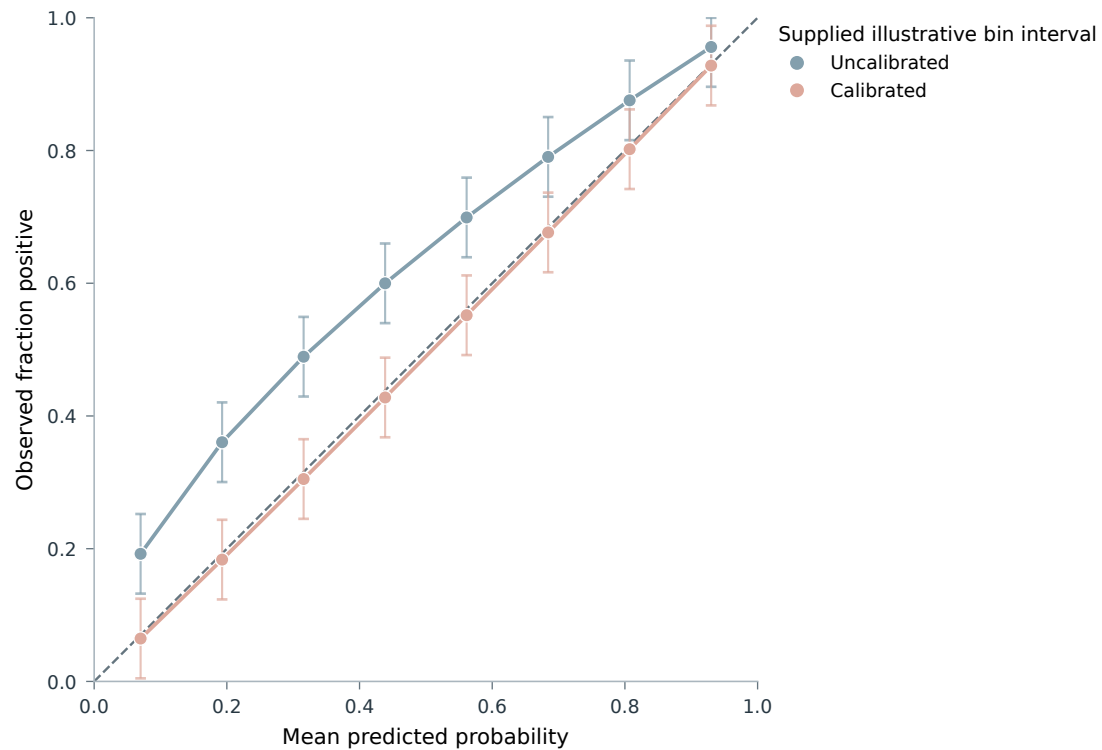
SYNTHETIC EXAMPLE

Precision-recall trade-offs



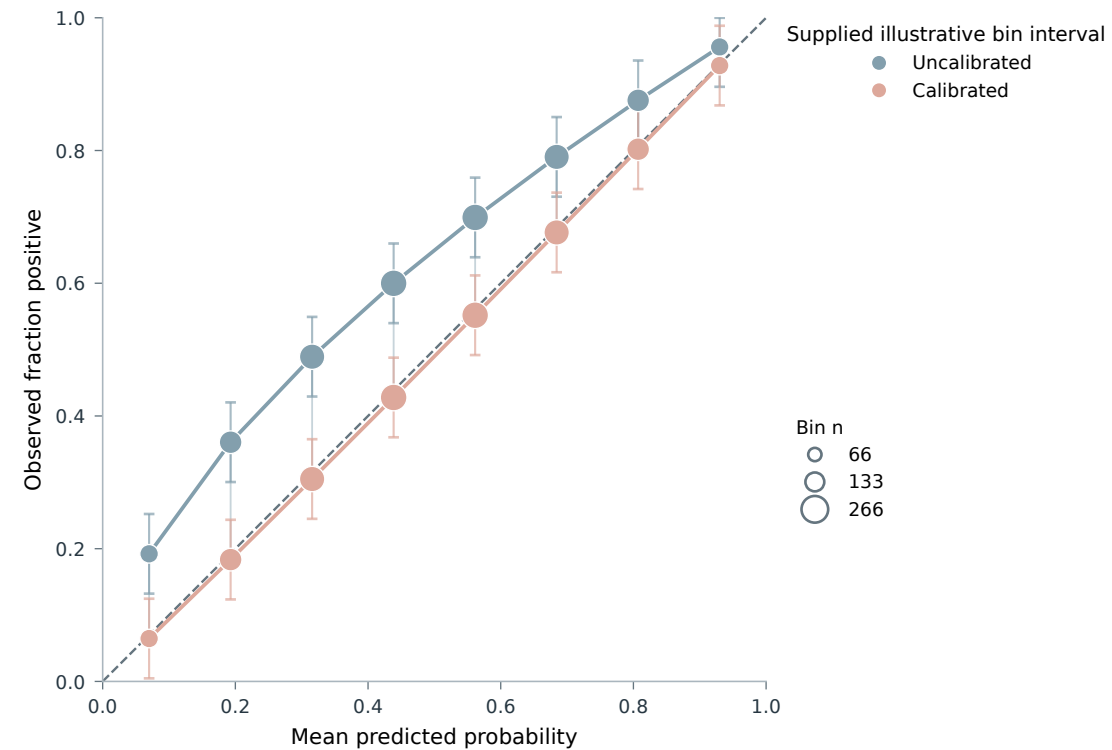
SYNTHETIC EXAMPLE

Probability calibration



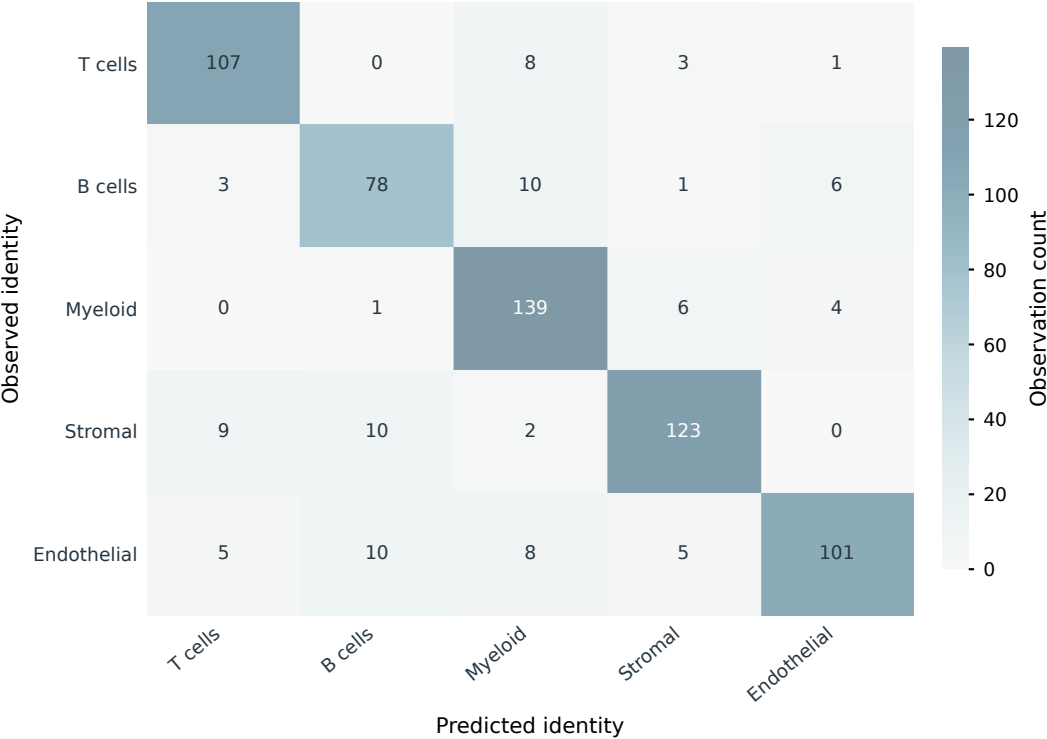
SYNTHETIC EXAMPLE

Probability calibration



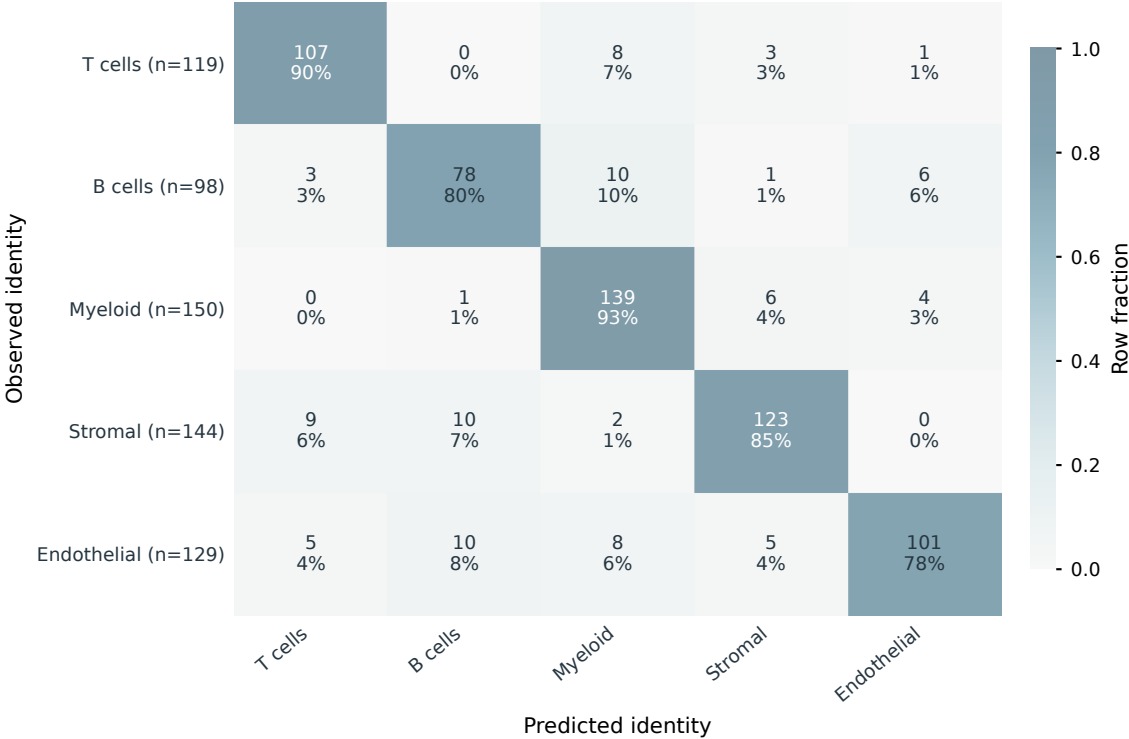
SYNTHETIC EXAMPLE

Classification errors by cell type



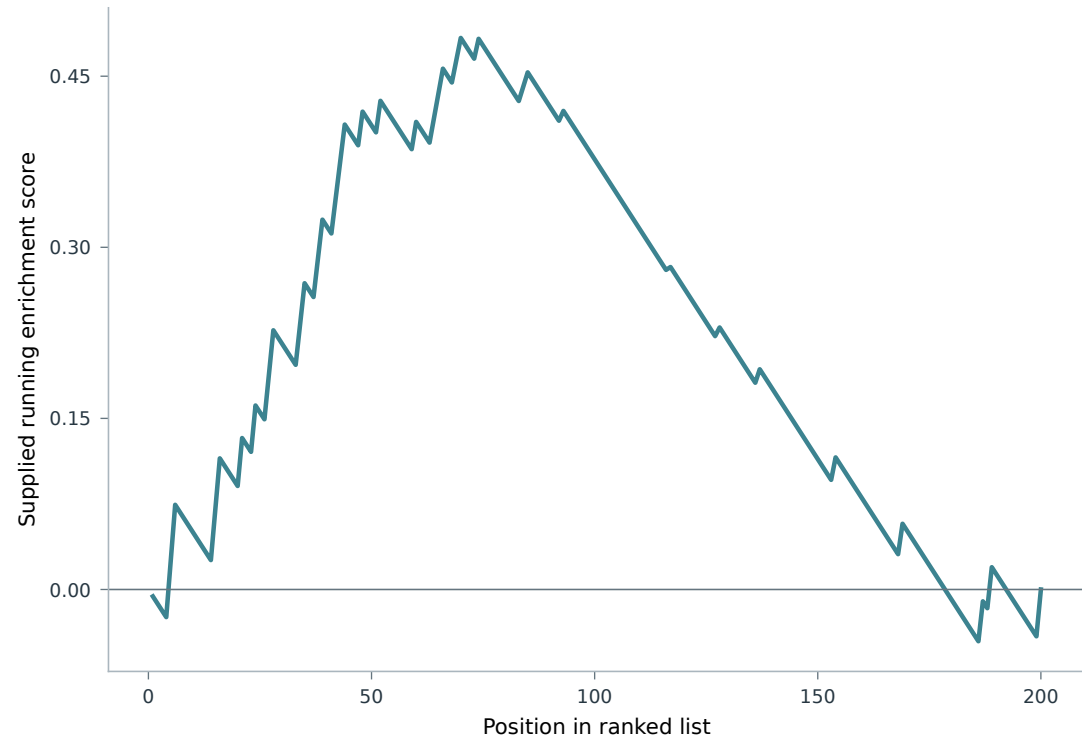
SYNTHETIC EXAMPLE

Classification errors by cell type



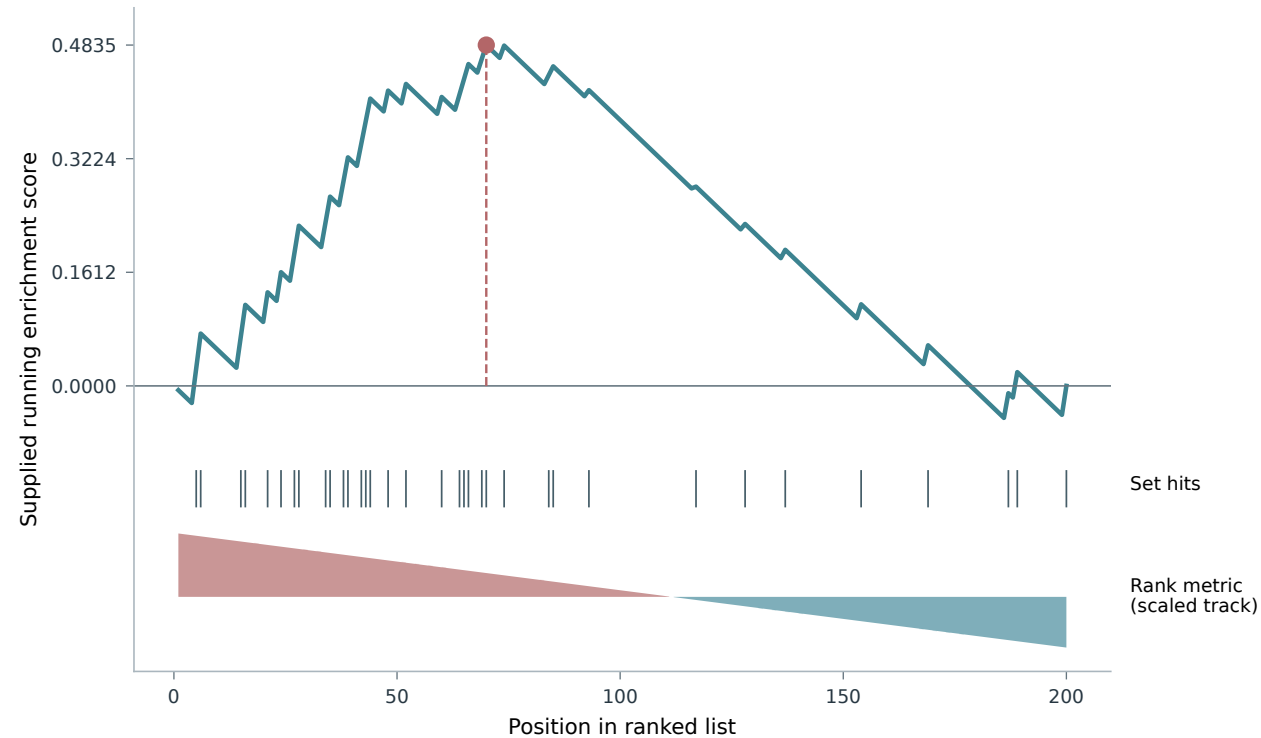
SYNTHETIC EXAMPLE

Ranked enrichment profile



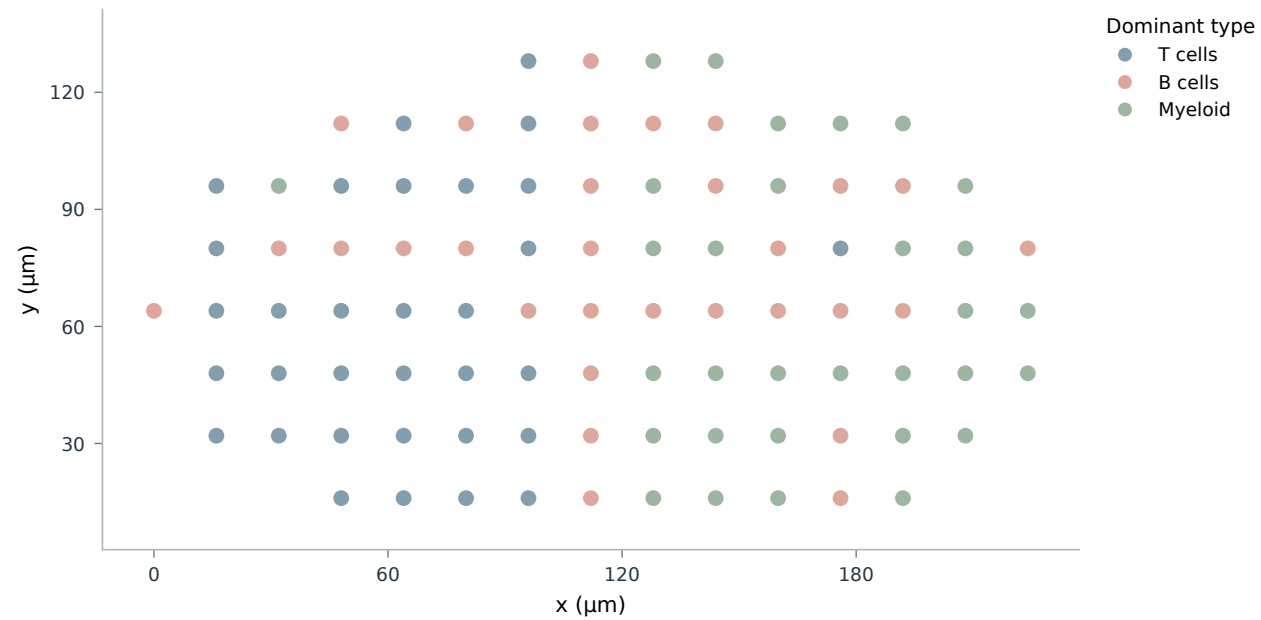
SYNTHETIC EXAMPLE

Ranked enrichment profile



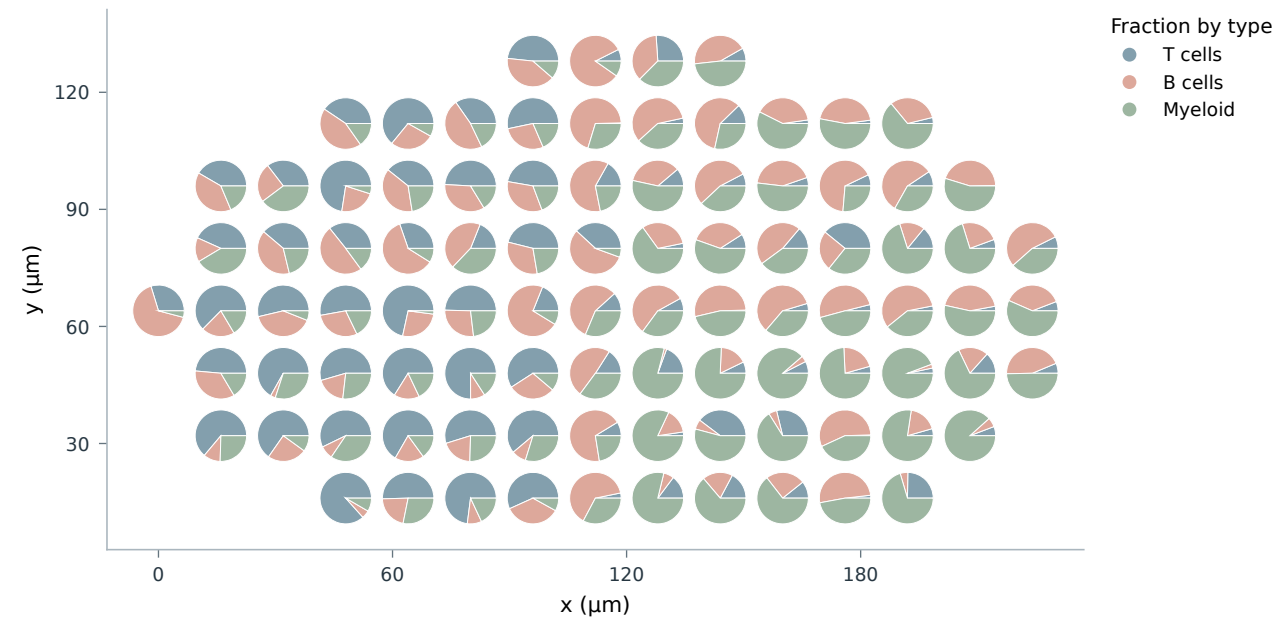
SYNTHETIC EXAMPLE

Spatial mixtures of cell identities



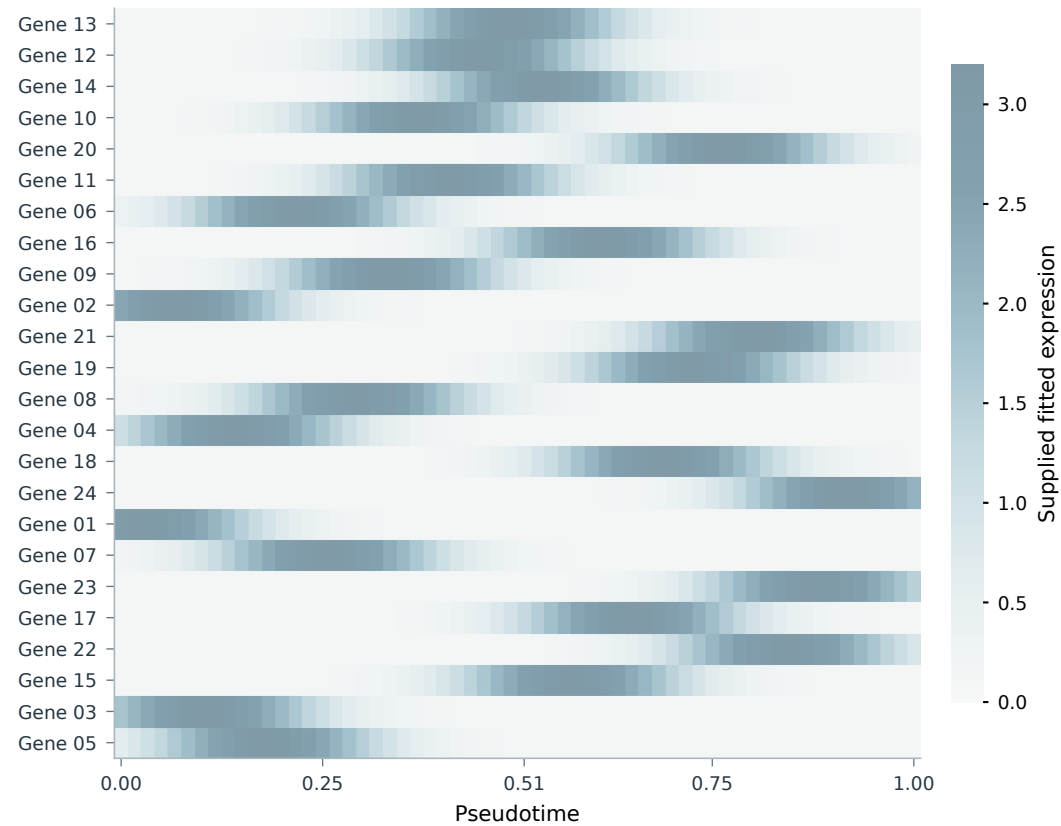
SYNTHETIC EXAMPLE

Spatial mixtures of cell identities



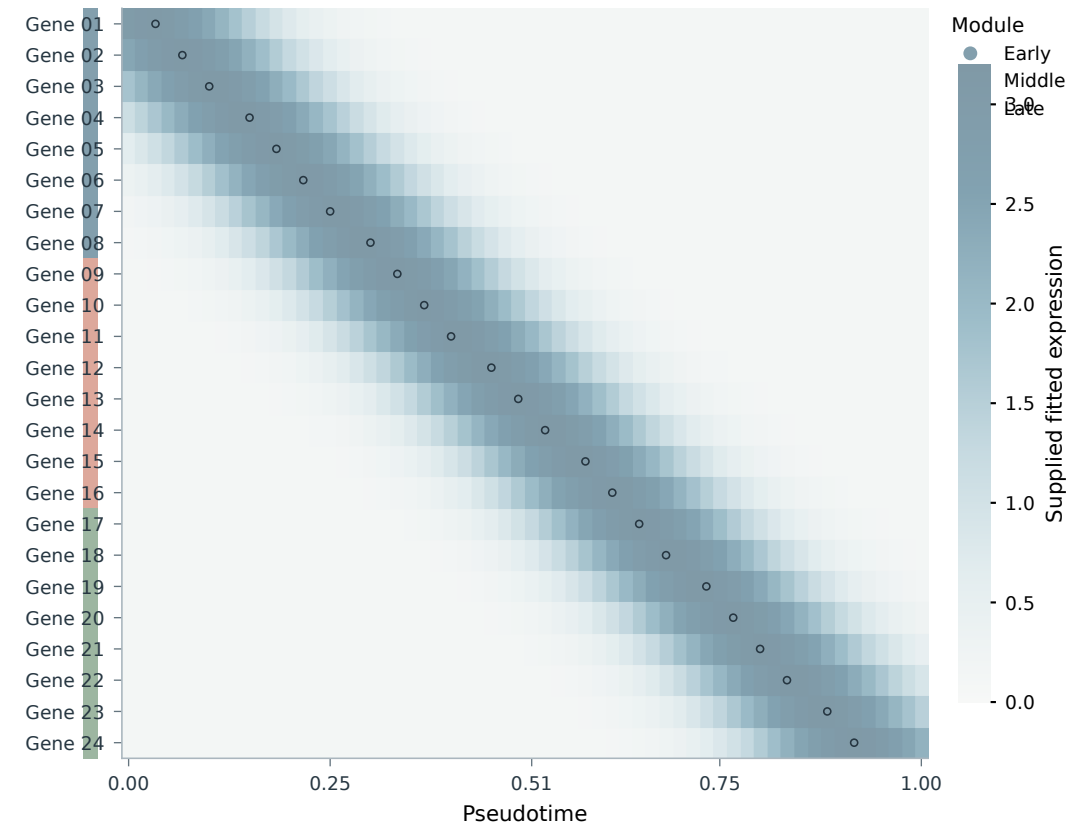
SYNTHETIC EXAMPLE

Temporal gene programmes



SYNTHETIC EXAMPLE

Temporal gene programmes



SYNTHETIC EXAMPLE